

Quantifying Racial Prejudices with Housing Transaction Data*

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Abstract

This paper aims to quantify racial prejudices using transaction-level housing market data. We pin down the impact of a marginal change of racial composition in a narrowly-defined neighborhood on the price appreciation between repeated sales of a house, and we find that an additional nonwhite household within a radius of 0.2 miles reduces the price appreciation by 1.27 percentage points. The effects are weaker in neighborhoods with a thicker housing market and a higher income level, suggesting that the role of racial prejudices is limited by market forces. The effects are also associated with voting behaviors and public anti-racism statements made on social media, indicating that they are likely to be driven by taste-based racial preferences. While we also find evidence of racial price discrimination effects as in the literature, these effects tend to be a form of statistical discrimination driven by other factors that correlate with race.

JEL Classification: J15, R23, R30, H00

Keywords: Housing market, repeated sales, racial prejudices, machine learning.

Data Availability: The data that support the findings of this study are available from Zillow but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available.

*We would like to thank Zillow for providing us access to the Zillow Transaction and Assessment Dataset (ZTRAX). More information on accessing the data can be found at <http://www.zillow.com/ztrax>. The results and opinions are those of the authors and do not reflect the position of Zillow Group.

1 Introduction

Housing is more than a consumption good; it involves neighborhood effects shaped by the behaviors and characteristics of residents. Preferences for quietness, community support, and safety often influence housing choices, alongside the neighborhood racial composition. Racial prejudices, i.e., preferences for or against certain racial groups, play a key role in these decisions. This paper quantifies such prejudice by analyzing how the price appreciation of a repeatedly transacted house is affected by changes in the racial composition of its immediate neighborhood using housing transaction data from the state of Virginia.¹

In contrast to numerous studies that ask a similar question, we define a neighborhood as an area that is much smaller than a census tract, the unit that most existing studies are based on. According to the U.S. Census Bureau, census tracts generally have a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. When people with racial prejudice are searching for a home, they should be more concerned with the racial composition of their immediate neighborhood rather than that of the entire census tract. Therefore, for each repeatedly transacted house, we define the neighborhood using a radius ranging from 0.2 miles at the low end to 1 mile at the high end. Each of our neighborhoods includes a few blocks at most, and the neighbors in it are people that one is likely to meet and interact on a frequent basis. Using this narrow definition of a neighborhood, we are able to accurately identify racial prejudices, with other relevant effects that are defined for larger areas removed by fixed effects and control variables.

We find strong evidence of racial prejudice at the neighborhood level. Even after accounting for economic status and time trends, house price appreciation between repeated sales is negatively and significantly affected by the entry of nonwhite households, while the effect of white households is positive but insignificant. In particular, we find that an additional nonwhite

¹We follow the literature to define racial prejudice and racial price discrimination differently. According to [Myers \(2004\)](#), “Prejudice is defined as an attitude about a group of people; it does not necessarily result in direct adverse action against that group. Discrimination, on the other hand, is some action or treatment of a group that differs from that received by others.” According to this definition, the change of home price appreciation due to changes in neighborhood racial composition is reflection of racial prejudice.

household within a 0.2-mile radius of a repeated transacted house reduces its price appreciation by 1.27 percentage points, which translates into a decrease in home value of \$4,800 due to ten extra nonwhite neighbors for a typical house of \$380,000 in the state of Virginia. Given that the inclusion of census tract by year fixed effects helps to account for the racial composition, and other characteristics such as school quality and crime rate, at the census tract level, we interpret our estimate as the impact on housing price appreciation of the racial composition of the immediate neighborhood relative to its census tract. We also find that this impact weakens as the neighborhood radius expands. When the neighborhood is expanded to a 1-mile radius, the impact of the neighborhood racial composition, relative to its census tract, diminishes to zero. Robustness checks and instrumental variable strategies confirm the validity of our findings.

To confirm that the observed effects are driven by racial preferences, we compare results across sub-samples. First, we divide neighborhoods by housing market thickness. In areas with more frequent transactions, residents who dislike their neighbors can more easily move out, and others can more easily move in. Consistent with Becker’s hypothesis that market forces reduce the impact of racial preferences ([Becker, 1957](#)), we find that the effects of racial prejudice are weaker in neighborhoods with thicker housing markets.

Second, we split the sample by buyer income. Since income is strongly correlated with education, and research shows that more educated individuals tend to display weaker biases against racial minorities ([Maykovich, 1975](#)), we expect differences in behavior. Consistent with this, we find stronger racial prejudice effects among lower-income buyers and significantly weaker effects among higher-income buyers.

Third, we examine whether racial preferences vary by political stance or public racial views. Since housing market decisions are largely private, individuals can act on racial preferences without making them explicit (one can move to a neighborhood that suits you, or move away from one that does not). We compare this with more public indicators—county-level voting outcomes and anti-racism statements on Twitter. Consistent with expectations, we find that racial prejudice effects are weaker in more Democratic-leaning counties and in areas with more

public anti-racism statements.

We contribute to the literature on housing market racial discrimination in several ways. First, we use large-scale transaction data from Zillow, capturing actual market values at the time of sale (rather than the homeowner’s self assessed value), allowing us to more accurately estimate racial prejudice effects. Second, the Zillow data allow us to define neighborhoods narrowly, focusing only on immediate neighbors who are likely to influence housing preferences, excluding less relevant households which comprise the majority in a census tract. This reveals strong evidence of racial prejudice, with price appreciation declining as more nonwhite households move in. Third, we distinguish taste-based racial preferences from statistical discrimination, indicating that the observed effects are primarily driven by the former. While we also find evidence of racial price discrimination effects, i.e., price differentials between non-white and white sellers/buyers (as in [Bayer et al. \(2017\)](#)), our results suggest that they are likely driven by other factors that correlate with race (e.g., statistical discrimination) rather than racial preferences.

The rest of the paper is structured as follows. Section 2 surveys the related literature. Section 3 describes the data sources and empirical methodology. Section 4 presents and discusses the empirical results, including robustness checks and exclusion of other mechanisms. Section 5 tests if the effects differ across races or change over time. Section 6 concludes.

2 Related Literature

Studies on racial discrimination in the housing market date back at least to the 1970s and this topic remains a central issue in the literature.² [Cutler et al. \(1999\)](#) argue as blacks move into white communities since the 1970s, institutional racism, such as redlining, racial steering,

²There is a large literature on the dynamics and formation of neighborhoods, explaining the integration and segregation by race in the US. Examples include [Bayer et al. \(2004\)](#), [Bayer et al. \(2014\)](#), and [Hwang and Sampson \(2014\)](#), and more recently [Bakens et al. \(2018\)](#), [Aliprantis et al. \(2022\)](#), and [Carrillo and Rothbaum \(2022\)](#). While these studies try to explain the location choices of people of different races, this paper only uses the movements as an explanatory variable.

lending discrimination, and land-use regulation, that enforce housing segregation,³ has been replaced by non-institutional (or decentralized) racial prejudice by market participants such as real estate agents, landlords, buyers, and sellers.⁴ While both cause housing segregation, there is a fundamental difference between racial price discrimination and racial prejudice (Myers, 2004). Many of the studies in the literature focus on the racial price discrimination, i.e., differential prices paid by white versus nonwhite buyers for comparable houses. For instance, many of the early studies including King and Mieszkowski (1973), Yinger (1978), and Schafer (1979) which use data from the 1960s find that blacks pay price premia in black and integrated areas. More recent studies which use better neighborhood (Chambers, 1992; Ihlanfeldt and Mayock, 2009) and/or repeated-sales data (Bayer et al., 2017) to address the omitted-variable bias also find that black and Hispanic buyers pay economically and statistically significant premia compared to white buyers. Instead, our paper focus on the racial prejudice effects and investigates the impact of changing racial composition of a small neighborhood in which a house is located on the value appreciation of the house.

A closely related study to ours is that of Moyer (2014). Using the 1990 and 2000 US census data to measure changes in racial composition, Moyer (2014) conducts a census tract level analysis on the Philadelphia housing market and finds that home value appreciation is negatively associated with racial turnover in predominantly white neighborhoods. Unlike Moyer (2014), we study the value appreciation of a house between two repeated sales and how it depends on the number of white and nonwhite households that move into the neighborhood in between the two sales using transaction level data. More importantly, our definition of neighborhood for a house is an area within a certain radius that varies from 0.2 to 1 mile rather than a census tract, which is too large to allow for a clean identification of racial prejudice

³Such evidence includes Ross (1971), Massey and Denton (1993), Squires (1994), Reibel (2000), Galster and Godfrey (2005), Crossney (2010), and Shertzer et al. (2016). Also see a literature review by Ellen and Ross (2018) on the causes and consequences of housing segregation, and a review by Glaeser et al. (2004) on John Kain's early contribution in this area, e.g., Kain (1968); Kain and Quigley (1970, 1972, 1975).

⁴Such evidence includes Ellen (2000), Card et al. (2008), Ellen et al. (2012), and Kollmann et al. (2018) among others.

effects.⁵

The first attempt of defining small neighborhoods to study discrimination and prejudice effects in the literature is made by [Myers \(2004\)](#). She uses survey data from the American Housing Survey (AHS) conducted by the U.S. Census Bureau. In 1985, 1989, and 1993, between 648 and 936 households were surveyed together with their 10 nearest neighbors within a one mile distance that are not separated by a bridge, river, railroad tracks, interstate highway, etc. The survey includes questions about the racial characteristics of the reference person, neighborhood quality, and either the respondent's report of monthly contract rent or the owner's self estimate of home value. Controlling for neighborhood characteristics, the paper finds that black owners pay premia of around 10 percent for housing and home values are negatively associated with the percentage of blacks in the neighborhood. While this paper is also motivated by the need of defining confined neighborhoods, it differs from [Myers \(2004\)](#) in a few ways. First, while she uses survey data of less than 1,000 households and their 10 nearest neighbors, we use repeated-sales data of 95,967 houses and all of their neighbors (with transactions in between the repeated sales) within a one-mile radius. Second, her paper measures home value using the owner's self estimate, which tends to overstate the value of the home ([Kiel and Zabel, 1999](#)). Even worse, the degree of price misperception has been found to significantly differ across

⁵Studies on racial disparities in housing returns are also related to our paper. For example, [Kim \(2000\)](#) studies differences in racial composition across census tracts in Milwaukee, Wisconsin and finds that, compared to neighborhoods resided by whites, those that are heavily resided in by racial minorities have experienced substantially lower rates of home price appreciation. [Macpherson and Sirmans \(2001\)](#) construct a county-level constant quality house price index for Tampa and Orlando, Florida and find that the change in percent African American has a negative effect on house price appreciation in both cities. By merging the Zillow's Home Value Index and Home Mortgage Disclosure Act data, [Kahn \(2021\)](#) finds that Blacks earn a lower rate of return on home purchases than Asians and Hispanics, due to the variation in racial composition across geographic locations. [Kermani and Wong \(2022\)](#) find that the racial gap in realized housing returns is due to a higher likelihood of distressed home sales, such as foreclosures and short sales, experienced by black and Hispanic homeowners. A second set of related studies focus on the impact of in-migration of certain races on housing market outcomes. In general, in-migration leads to an increase in housing demand and therefore increased housing/rental prices (see [Saiz \(2003\)](#); [Gonzalez and Ortega \(2013\)](#); [Howard \(2020\)](#)). However, as [Saiz and Wachter \(2011\)](#), [Akbar et al. \(2022\)](#) and [Daeppe et al. \(2023\)](#) show, nonwhite in-migration is associated with lower housing price appreciation. Another strand of the literature that is related to our paper studies the impacts of neighborhood effects on house prices. These neighborhood effects include environmental goods ([Boyle and Kiel, 2001](#)), renovations and maintenance of neighbors' houses ([Rossi-Hansberg et al., 2010](#); [Ioannides, 2002](#)), schools ([Bayer et al., 2007](#)), short-term rental regulation ([Kim et al., 2017](#)), and neighbors' income and how similar they are to the homeowners ([Ioannides and Zabel, 2003, 2008](#); [Leung and Tsang, 2012](#)).

racial groups ([Carrillo et al., 2023](#)). Instead, we use actual sale prices from repeated transaction data, which mitigate unobserved heterogeneity. Third, her paper estimates the effects of the racial composition of a neighborhood at a given point in time on home value, and therefore quantifies the cross-sectional aspect of racial prejudice effects. In contrast, we use the number of white and nonwhite households that move into a neighborhood to capture the change in the racial composition over time and study its impact on the sale prices of repeatedly transacted houses, adding a time dimension to quantifying racial prejudice effects.

3 Data and Methodology

3.1 ZTRAX and HMDA data

The main data are obtained from the Zillow Transaction and Assessment Databased (ZTRAX) and the Home Mortgage Disclosure Act (HMDA). The ZTRAX provides more than 20 years of digitized county records of all information relevant to a home transaction, including the price, home characteristics, geographic information, and loan details across more than 2,750 U.S. counties. The HMDA requires financial institutions to report and disclose loan-level information about mortgages, including the loan amount and type, applicant’s demographic information, and geographic information.

Our sample is restricted to the state of Virginia between 2007 and 2020. We choose Virginia for several reasons. First, Virginia is one of the few states with a reasonable portion of housing transactions that can be matched with the mortgage data. Second, Virginia has large variations in both racial composition and political stance across geographical regions. According to the 2020 U.S. Census results, the percentage of white population was higher than 95% in Dickenson County, Russell County, and Highland County, but lower than 35% in Petersburg City, Emporia City, and Manassas Park City. The 2020 U.S. presidential election results show that the Democratic candidate received more than 80% of the votes in Petersburg City, Charlottesville City, and Richmond City, but less than 16% in Lee County, Bland County, and Scott

County. Compared to other states, Virginia is not as diverse as California, Nevada, Texas, New York, or Florida, but more diverse than most of the North West and Mid-West states. In 2020, Virginia has a Diversity Index (the chance that two people chosen at random are from different racial groups) of 60.5%, which lies in the second highest tier.

We first drop observations with missing or zero sale price or loan amount and those with missing information on census tract or transaction year in the ZTRAX data. Non single-family residential properties are also excluded. We then merge the ZTRAX transaction data with HMDA mortgage data based on the transaction year, loan amount, and census tract. A total number of 705,151 observations are matched along these three indicators to the HMDA data and 920,701 observations cannot be matched. Among the matched observations, 556,404 are uniquely matched on the three variables. For those that are not uniquely matched, we add the lender's name as an additional indicator using Stata's "matchit" command with the "soundex" algorithm. This adds another 35,030 observations to our sample pool with a total of 591,434 observations. We refer to this as our transaction sample. The top panel of Table 1 presents summary statistics of sale price and applicant income in the transaction sample. It is worth noting that both sale price and applicant income are reported in millions of dollars to avoid having very small coefficient estimates in the regression tables.

[Table 1 about here.]

We construct our repeated-sales sample by including the properties in the transaction sample that have been transacted at least twice. If a property is transacted more than twice, the same property shows up multiple times and each pair of two consecutive transactions is listed as an observation. We further exclude properties that have been remodelled between the two sales (less than 1%) due to lack of details about the remodelling.⁶ This gives rise to our repeated-sales sample with 95,967 properties, each has been transacted twice, in 1,600 census tracts in the state of Virginia. It is important to note that properties in the repeated-

⁶The reason why remodelling accounts for a very small portion of the observations is perhaps that remodelling in the ZTRAX assessment data pertains to major renovations that require permits from the county.

sales sample are somewhat less expensive than those in the transaction sample; see Table 15 in the appendix. Although this difference is statistically significant, it does not result in any meaningful economic disparity between the two samples.

We are interested in relating the percent price appreciation between two consecutive transactions on a property to the buyer and seller races, and changes in the racial composition of neighbors within a certain distance from the property, controlling for a series of transaction, location, and time characteristics. To account for changes in the racial composition in a neighborhood, we calculate the distance between the repeatedly transacted property and each of the properties in the transaction sample sold in a year between the two transactions of the repeatedly transacted property using the latitude and longitude data. We use the number of properties within a certain distance from the repeatedly transacted property purchased by white households and nonwhite households to capture changes in the racial composition of the neighborhood. A neighbor is defined as a white household if the sole applicant of its mortgage is white or both the applicant and co-applicant are white. Otherwise, the neighbor is classified as a nonwhite household. It is worth noting that, following Myers (2004), the white vs. non-white classifications are strictly based on race, and not on ethnicity. According to this definition, 405,666 homebuyers (about 68.6%) in our transaction sample are whites and 108,802 homebuyers (about 18.4%) are nonwhites, and the rest have missing information on race. Among the 108,802 nonwhite homebuyers defined jointly by the applicant and co-applicant race, about 7% of the applicants are White, 2.5% American Indian or Alaska Native, 29% Asian, 59% Black or African American. Figure 1 presents an illustrative map where the pentagram denotes a property in the repeated-sales sample in Roanoke City, VA, sold in 2014 and 2016, with its neighborhood of a radius of one mile, denoted by the circle. The squares and diamonds represent white and nonwhite households, respectively, that move into the neighborhood between 2014 and 2016, exclusively. In this illustrative example, seven white households and seven non-white households move into the neighborhood of the repeatedly transacted property between its two repeated sales.

[Figure 1 about here.]

It is worth noting that the data we use have several limitations. First, while we use the race of the household that buys a house in the neighborhood to capture changes in its racial composition, we do not have enough data to identify the race of the seller. In other words, we only have information about who moves in but not who moves out. In order to identify the seller's information, we need to use the repeated-sales sample rather than the transaction sample to count the number of households that move into the neighborhood, which would leave out a large part of the changes that happen in the neighborhood. While a majority of white buyers, about 89%, buy their houses from white sellers, about 72% of nonwhite buyers also buy houses from white sellers. This implies that white relative to nonwhite purchases would overstate the true changes in the white relative to nonwhite composition in a neighborhood. As a result, our estimated marginal racial prejudice effects would understate the true effects of changes in racial composition. To mitigate this problem, later in Section 4 we conduct a robustness check using a race prediction algorithm to predict the seller's race in the transaction sample.

Second, given that our transaction sample is obtained from merging the ZTRAX transaction data with the HMDA mortgage data, we do not have information about those who buy a house in the neighborhood without having a mortgage. According to the Realtors Confidence Index Survey, the share of cash sales varies between 12% and 35%, with an average of around 25%. Suppose there is no systematic difference in the number of cash offers between white and nonwhite buyers, our estimated marginal racial prejudice effects would equally overstate the true racial effects of nonwhite and white neighbors by about one third. If white buyers are more likely to make cash offers, our estimated racial prejudice effects would overstate the true racial effects of nonwhite neighbors by less than one third and those of white neighbors by more than one third.

3.2 Presidential election data

The 2020 presidential election data are directly obtained from the Virginia Department of Elections.⁷ We use the percentage of votes received by the Democratic presidential candidate by locality (i.e., county) to capture the geographic variation in political stances. The summary statistics of the election data are presented in Table 2.

[Table 2 about here.]

3.3 Twitter data on #BlackLivesMatter

We follow Boxell and Conway (2022) and Leung and Strumpf (2022) to adopt a machine learning approach to construct a measure of online racial preference using data from social media. To do that, we collect two sets of Twitter data. First, to construct a labeled dataset on racial preference for training our model, we use the Twitter API to extract all tweets containing the hashtags #BlackLivesMatter, #AllLivesMatter and #BlueLivesMatter between 2014 and 2021. For each tweet, we extract the tweet content, the date and time of the tweet, the number of likes, and the number of retweets. Initially, we obtain slightly more than 10 million tweets, majority of them with the hashtag #BlackLivesMatter (8.6 million, or 83% of total), and the rest of them with either #AllLivesMatter or #BlueLivesMatter (1.7 million). We then apply the standard pre-processing steps to all tweets including lemmatization, punctuation, removals of hastags, URLs, stop-words, and non-English words. After dropping tweets with too few strings (less than four words) and randomly dropping portion of the tweets with #BlackLivesMatter to create a balance between the two types of hashtags, we end up with a database of close to 2.6 million tweets, in which tweets containing #BlackLivesMatter composes roughly half of the sample.

Second, we collect all the geo-tagged tweets sent out from Virginia in 2020. Again, in addition to the exact location at the census tract level, we extract the tweet content, the date

⁷<https://www.elections.virginia.gov/resultsreports/election-results/>

and time of the tweet, the number of likes, and the number of retweets for each tweet. We end up with approximately 5.7 million tweets in this dataset.⁸

From the first set of data containing the hashtags #BlackLivesMatter, #AllLivesMatter and #BlueLivesMatter, we use 2/3 of the sample as training data and the remaining 1/3 as testing data. In our application, we use bigrams, which are two-word combinations, as the unit of words and phrases, which yields a better prediction than using unigrams (i.e., single words). To account for the importance of each word, we apply the Term Frequency-Inverse Document Frequency (TFIDF) model, which is the product of the frequency of a word (term frequency, TF) and the importance of the word (inverse document frequency, IDF), in the vectorization of tweets. We then use the Linear Support Vector Classification (Linear SVC) model in the scikit-learn package, a Python module, as our machine learning model.

Overall, the model achieves 75% accuracy on the leave-out validation dataset. In addition to this, we also construct the confusion matrix to evaluate the goodness of fit of the model. Figure 2 shows that our model can predict both types of hashtag correctly 75% of the time, with 25% chance of wrongly labelling the tweets of the other hashtag.

[Figure 2 about here.]

We then apply the model to “predict” the hashtag of the Virginia Twitter data. The confidence of the prediction (in terms of probability of it with a hashtag of #BlackLivesMatter) is our measure of racial preference. To construct the county-level BLM score, we first take the average BLM score for each Twitter user and then add up the scores across users in each county. The county-level total score is then scaled by the voting age population (in thousands). We therefore avoid counting any frequent Twitter user’s racial view multiple times. The summary statistics of the BLM score variable are also presented in Table 2.

⁸Sloan et al. (2013) demonstrate that only approximately 0.85% of tweets are geo-tagged as a result of the opt-in policy of Twitter, which raises the question of whether the opt-in sample is representative of the whole Twitter world. Indeed, studies have shown that both the users’ profiles (Sloan and Morgan, 2015) and their tweet content (Karami et al., 2021) are not representative of the Twitter population. Having said that, to the extent that the opt-in behavior does not vary significantly across counties in Virginia, our measure should still reflect the geographical variation of the racial preference. As we will show in Table 2, our measure and the Biden vote percentage are highly correlated.

3.4 Methodology

Our primary approach follows that of [Bayer et al. \(2017\)](#), which relates the log price of transaction for house i in neighborhood n of census tract c at time $t = t_0, t_1$ to the race of both the seller and buyer, and a set of control variables:

$$\begin{aligned} \ln(\text{Price}_{i,n,c,t}) = & \alpha \text{NonwhiteSeller}_{i,t} + \beta \text{NonwhiteBuyer}_{i,t} + \gamma \text{BuyerChars}_{i,t} \\ & + \phi \text{HouseChars}_i + \theta \text{NeighborhoodChars}_{n,t} + \mu_{c,t} + v_{i,c,t}, \end{aligned} \quad (1)$$

where “NonwhiteSeller” is a dummy variable that equals one if the seller is nonwhite (i.e., American Indian or Alaska Native, Asian, Black or African American, or Native Hawaiian or Other Pacific Islander) and zero otherwise, and “NonwhiteBuyer” is a dummy variable that equals one if the buyer is nonwhite and zero otherwise. The parameter α therefore measures how much a nonwhite seller receives relative to a white seller when selling, i.e., the price differential between nonwhite and white sellers, and β measures how much a nonwhite buyer pays relative to a white buyer when buying, i.e., the price differential between nonwhite and white buyers. These two parameters jointly measure racial price discrimination, which can be driven by a negative α and/or a positive β .

It is worth noting that our race information is obtained from the homebuyer’s loan application record. Therefore, the seller’s race is not observed for the very first transaction in our repeat-sales sample; for the second transaction, it is equal to the buyer’s race for the first transaction (and so on for the third etc.). The buyer’s race is defined jointly based on the loan applicant and co-applicant (if exists). In particular, the buyer is considered white if either the only loan applicant is white or both the applicant and co-applicant are white. Otherwise, the buyer is considered nonwhite. Following [Bayer et al. \(2017\)](#), “BuyerChars” include the buyer’s characteristics other than his or her race, such as income and debt-to-income (DTI) ratio, and the squared terms of these variables. “HouseChars” are time independent house characteristics. “NeighborhoodChars” include the racial composition of the narrowly defined neighborhood.

The term $\mu_{c,t}$ captures census tract by time fixed effects, and $v_{i,c,t}$ is the error term.

Taking the difference of log price between time t_1 and t_0 yields:

$$\begin{aligned} \Delta \ln(\text{Price}_{i,n,c,t_0,t_1}) = & \alpha \Delta \text{NonwhiteSeller}_{i,t_0,t_1} + \beta \Delta \text{NonwhiteBuyer}_{i,t_0,t_1} + \gamma \Delta \text{BuyerChars}_{i,t_0,t_1} \\ & + \theta \Delta \text{NeighborhoodChars}_{n,t_0,t_1} + \mu_{c,t_1} - \mu_{c,t_0} + v_{i,c,t_1} - v_{i,c,t_0}, \end{aligned} \quad (2)$$

where $\Delta \ln(\text{Price}_{i,n,c,t_0,t_1}) = \ln(\text{Price}_{i,n,c,t_1}) - \ln(\text{Price}_{i,n,c,t_0})$, which measures the nominal price appreciation of the house i bought in year t_0 and sold in year t_1 in neighborhood n of census tract c . Using nonwhite and white households that move into the neighborhood n between t_0 and t_1 , exclusively, to capture changes in the neighborhood attributes (and controlling for their economic status), Equation (2) reduces to:

$$\begin{aligned} \Delta \ln(\text{Price}_{i,n,c,t_0,t_1}) = & \alpha \Delta \text{NonwhiteSeller}_{i,t_0,t_1} + \beta \Delta \text{NonwhiteBuyer}_{i,t_0,t_1} + \gamma \Delta \text{BuyerChars}_{i,t_0,t_1} \\ & + \theta_1 \text{Nonwhite move-ins}_{n,t_0,t_1} + \theta_2 \text{White move-ins}_{n,t_0,t_1} \\ & + \theta_3 \text{Avg income of move-ins}_{n,t_0,t_1} + \mu_{c,t_0} + \mu_{c,t_1} + v_{i,c,t_0,t_1}, \end{aligned} \quad (3)$$

where “Avg income of move-ins $_{n,t_0,t_1}$ ” is the average income of those households, either white or nonwhite, that bought a house in the neighborhood n of house i between t_0 and t_1 . While α and β capture racial price discrimination, racial prejudice effects, if exist, are captured jointly by the parameters θ_1 and θ_2 . To the extent that Nonwhite move-ins $_{n,t_0,t_1}$ and White move-ins $_{n,t_0,t_1}$ are not endogenously affected by omitted variables and reverse causality, θ_1 measures the premium that home buyers are willing to pay for one less nonwhite neighbor and θ_2 measures the premium that home buyers are willing to pay for an additional white neighbor.⁹ Summary statistics of all model variables in Equation (3) can be found in the bottom panel of Table 1. In particular, the average price appreciation is around 13% across 95,967 repeated sales. Using the 0.2-mile radius definition of a neighborhood, there are 4 nonwhite and 14 white house-

⁹Section 4.2 tests for these potential concerns. While we do not completely rule out the possibility of endogeneity, our results in the section suggest that our main results are not likely to be biased by endogeneity. We also consider two sets of instruments for nonwhite and white move-ins and obtain similar results.

holds that move into the neighborhood between repeated sales of a property on average in the repeated-sales sample. These figures increase as we expand the neighborhood. Using the 1-mile radius definition, for instance, there are 51 nonwhite and 183 white move-ins between repeated sales on average.

In our repeated-sales sample, some houses are transacted twice and some others are transacted more than twice. Since we do not observe the seller’s race for the first transaction of each house, the variable “ $\Delta\text{NonwhiteSeller}$ ” is also not defined in this case. Following [Bayer et al. \(2017\)](#), we define a dummy variable “Missing $\Delta\text{NonwhiteSeller}$ ” that equals one if “ $\Delta\text{NonwhiteSeller}$ ” is missing and zero otherwise. Then we set the missing values of “ $\Delta\text{NonwhiteSeller}$ ” to zero and augment Equation (3) with the “Missing $\Delta\text{NonwhiteSeller}$ ” indicator. This allows us to estimate other parameters using the entire repeated-sales sample and use only the variation in “ $\Delta\text{NonwhiteSeller}$ ” from houses that have been transacted more than twice to identify the parameter α . Similarly, we also augment Equation (3) with a “Missing $\Delta\text{NonwhiteBuyer}$ ” indicator if the “ $\Delta\text{NonwhiteBuyer}$ ” is missing due to missing race information in the HMDA data and a “Missing avg income of move-ins” indicator if “Avg income of move-ins” is missing due to the absence of move-ins.

While it is impossible to control for all factors that may affect the housing price appreciation, we should control for important factors like school quality and crime rate, which may be time-varying. To capture these neighborhood characteristics and other factors that cannot be observed, we use time-specific fixed effects at the census tract level, i.e., μ_{c,t_0} and μ_{c,t_1} . We choose not to use fixed effects at a finer level (e.g., Census block group) to avoid selection bias: most observations will be dropped, and only block groups that have frequent transactions will remain in the sample. Typically, there are more census tracts than schools within a county. For instance, in Fairfax County, the most populated county in Virginia, there are 222 schools (according to the US News report on Fairfax County Public Schools) but 235 census tracts in our sample. Thus, census tract by year fixed effects are sufficient to capture time-varying school quality. Similarly, the concept of crime rate does not apply to a localized area; crime rate data is

never gathered at a finer level than ZIP codes. In Fairfax County, there are about 75 ZIP codes, which is fewer than the number of census tracts. Therefore, using census tract by year fixed effects effectively captures the time variation in school quality and crime rate, which are critical local factors influencing housing price appreciation. These fixed effects also help account for differential price appreciations in various census tracts during alternative time periods and for alternative holding periods. By including these fixed effects, we allow for differential price appreciations between any two time periods (t_0, t_1) and (t'_0, t'_1) and between any two census tracts c and c' . The regression therefore enables us to pick up the impact of nonwhite and white move-ins within a narrowly defined neighborhood on the net-of-tract housing price appreciation of the neighborhood.

4 Results and Interpretations

4.1 Baseline racial prejudice effects

Table 3 reports the baseline results for racial prejudice effects from the estimation of Equation (3) where an immediate neighborhood is defined as a 0.2-mile radius of a repeatedly transacted house. The dependent variable is multiplied by 100 so that the price appreciation between repeated sales is expressed in percent. In Column (1), we start with a simple specification that explains the price appreciation of a repeatedly transacted house to the number of nonwhite and white households that move into the neighborhood between the repeated sales. These two variables jointly capture the racial prejudice effects. Column (2) adds race changes of the buyer and seller to capture the racial price discrimination effects. Column (3) controls the average income of the households that move into the neighborhood and Column (4) further controls changes in the buyer's income and its square and changes in the buyer's DTI (calculated as the mortgage loan amount divided by the applicant's income) and its square.

The goodness-of-fit of the most comprehensive model specification is higher than 0.4. As expected, other things being equal, there exists an inverted U-shaped relationship between

housing price and the buyer’s income and an inverted U-shaped relationship between housing price and the buyer’s debt-to-income ratio. Controlling for these income and leverage factors, as well as changes in racial composition, we find that nonwhite sellers receive 1.379% (statistically significant) less than their white counterparts when selling and pay 0.112% (statistically insignificant) more than their white counterparts when buying. These two values indicate a roughly 1.5% racial price discrimination between nonwhite and white homeowners, which is close to the 2% racial price discrimination estimated by [Bayer et al. \(2017\)](#) from four major metropolitan housing markets.

[Table 3 about here.]

As shown in Table 3, in general, homeowners experience a financial loss as nonwhite households move into the neighborhood in very close proximity (0.2 miles) but do not necessarily benefit from white move-ins. The magnitude of the negative effects of a nonwhite neighbor, however, decreases as we add more controls. For instance, when the income of move-ins are controlled for, the negative effect of an extra nonwhite neighbor goes down slightly from 0.181 to 0.163 percentage points (from Column (2) to Column (3) in Table 3), suggesting only a small portion of the effect can be explained by the economic status of those neighbors. Also, the negative effect further goes down to 0.127 percentage points when the buyer’s income and leverage are controlled for (from Column (3) to Column (4) in Table 3), which suggests that the preference against nonwhite neighbors is at least partly, though not entirely, explained by the buyer’s economic status. These racial prejudice effects translate into a decrease, for a typical house of \$380,000 in Virginia (the average transaction price in 2020), in home value of \$4,800 for every ten extra nonwhite neighbors. It is worth noting that the racial prejudice effects are mostly driven by black households that move into the neighborhood.¹⁰

¹⁰Blacks account for the majority of nonwhite move-ins. We have also decomposed the nonwhite move-ins into black move-ins and other nonwhite move-ins, and we find that the former significantly lowers the housing price appreciation between repeated sales. Every ten extra black neighbors in the immediate neighborhood are associated with a decrease in home value of more than \$6,700 for a typical house in Virginia. These additional results are available upon request. We choose to define a nonwhite catchall group to capture all minority categories throughout this paper because, as we described in the data section, only 18.4% of move-ins are nonwhites and disaggregating the minority categories would result in certain categories having a lot of zero move-ins.

Given that racial prejudice effects reflect people’s taste for or against neighbors of certain races, they should care less about neighbors far away from them as they will encounter and interact with such neighbors less often. In Table 4, we use the most comprehensive model specification in Table 3 and vary the radius of a neighborhood from 0.2 miles to 1 mile in Columns (1) to (5).¹¹ Our estimates of the racial prejudice effects indeed diminish with the distance from the house. In particular, the marginal effects of nonwhite move-ins are estimated to be -0.0496 , -0.0321 , and -0.0156 , respectively for a 0.4-mile, 0.6-mile, and 0.8-mile radius neighborhood. These values translate into a decrease in home value of \$1,900, \$1,200, and \$600, for a typical house of \$380,000, due to the move-ins of ten additional nonwhite households. As the neighborhood is expanded to include all neighbors within a 1-mile radius of a house, the marginal racial prejudice effects shrink to 0.008 percentage points and become insignificant for nonwhite move-ins while white move-ins stay insignificant across all columns. This evidence also supports our idea of defining a narrow neighborhood to study racial prejudice effects.

[Table 4 about here.]

4.2 Endogeneity concerns

In this subsection, we test for two major sources of potential endogeneity, i.e., omitted variables and reverse causality. The tests suggest that our main results in Table 4 are not likely biased by endogeneity. Nevertheless, we further consider two sets of instruments for nonwhite and white move-ins and obtain similar estimation results.¹²

¹¹We use various neighborhood definitions from 0.2 to 1 mile of a distance instead of grouping neighbors into $(0, 0.2]$, $(0.2, 0.4]$, $(0.4, 0.6]$, $(0.6, 0.8]$, and $(0.8, 1]$ mile categories to avoid multicollinearity, given that the number of households that move into various distance categories are highly correlated.

¹²As pointed out by an anonymous referee, despite these checks, it is impossible to fully overcome endogeneity concerns. In other words, there might be some other unobserved factors at the immediate neighborhood level that affect housing prices but we fail to account for. There might be self selection of moving into certain neighborhoods by white and nonwhite households that we fail to detect. Nevertheless, we conduct a series of robustness checks and sub-sample analyses to show that what we identify is indeed racial prejudice.

4.2.1 Omitted variables

While our estimation has utilized a sample of repeated sales and taken into account the buyer's income and debt-to-income ratio as control variables and a rich set of fixed effects, it is still possible that other unobserved factors bias our estimates. To assess the scale of such bias, we adopt the sensitivity analysis method proposed by [Oster \(2019\)](#) to evaluate how strong selection on unobservables has to be in order to explain away our estimates. At the bottom of Table 4, we present the δ estimate of the proportional bias due to unobservables that would have to exist to drive the coefficient of nonwhite move-ins to zero. Therefore, a value of δ close to zero is an indication of non-robust estimates. On the contrary, a δ value far away from zero is an indication of robust estimates. In Columns (1) to (4) where nonwhite move-ins are estimated to reduce housing appreciation in a diminishing fashion, the δ estimate ranges between 1.785 and 8.787, suggesting that the omitted variables have to be 1.785 to 8.787 times as important as the included controls to explain away the negative coefficient. In other words, our estimates are considerably robust.

4.2.2 Reverse causality

One can argue that neighborhoods with lower housing price appreciations may attract more nonwhite buyers relative to white buyers. To address this reverse causality concern, we keep using the repeated-sales sample. Instead of counting the number of white and nonwhite move-ins between repeated sales, we calculate the number of white and nonwhite households that move into the neighborhood of a certain radius within one year after the second sale. We then regress the share of one-year ahead nonwhite move-ins (in percent) against the annualized price appreciation (in percent) to test for reverse causality. Results presented in Table 5 suggest that annual price appreciation is not a significant predictor of future nonwhite move-ins.

[Table 5 about here.]

4.2.3 Instrumental variable estimation

We have shown that our estimation results are not likely biased by omitted variables and reverse causality, two major sources of endogeneity. Nevertheless, we provide further support for our results using an instrumental variable approach in this subsection. In particular, we consider two sets of instruments for nonwhite and white move-ins.

The first set of instruments share some similarities with the shift factor of shift-share or [Bartik \(1991\)](#) instruments (but we do not have a share factor). We instrument the racial composition of a small neighborhood around a repeatedly transacted property with the racial composition of a broader neighborhood that appears to be irrelevant to the housing price appreciation. Since the nonwhite and white move-ins within the 1-mile radius are shown to have insignificant effects on the housing price appreciation between repeated sales (see Column (5) of Table 4), after controlling for time-specific census tract fixed effects, we use them as instruments for the nonwhite and white move-ins in the small neighborhood within the 0.2- to 0.8-mile radius. The logic behind the instrument is that, while changes in racial composition in the broader area do not affect house price directly, they are strongly correlated with changes in racial composition in a small neighborhood within that broader area.¹³

The second set of instruments exploit the racial composition of nearby neighborhoods. We instrument the nonwhite and white move-ins (within a certain distance of a repeatedly transacted property) with the average nonwhite and white move-ins (within the same distance of other repeatedly transacted properties located in the same census tract between the same two years of repeated transactions). One issue is that nearby neighborhoods within the same census tract may share the same housing trend. However, such effect has been captured by the year of transaction interacted with census tract fixed effects.

The IV estimation results are presented in Table 6, with the results using the first set

¹³For example, in the immigration literature, immigrants choose where to settle when they migrate to the U.S. and therefore immigration in a local area is considered endogenous. The widely used shift-share instrument for the local-area immigration is based upon a shift factor that captures immigration in a much broader area, such as the entire U.S. In our context, we cannot expand the broad area to include the entire census tract because we have already included time-specific census tract fixed effects in our regressions.

of instruments in the top panel and those using the second set of instruments in the bottom panel. First stage regressions are presented in Table 16 in the appendix, and the instruments are strongly relevant. As expected, the use of instrumental variables reduces the efficiency of the estimator. The standard error estimates are therefore larger than those of the baseline estimates in Table 4. Nevertheless, nonwhite move-ins still impact housing price appreciation negatively and the impact diminishes with distance.

[Table 6 about here.]

4.3 Robustness checks

4.3.1 Excluding non-individuals

Housing transactions can sometimes involve non-individual buyers or sellers. We retain these observations in our main analysis because our focus is on repeated transactions, where each observation includes two transactions and, for each transaction, two parties, i.e., a seller and a buyer. If we were to restrict the sample to transactions where all four parties are individuals, we would lose more than half of the observations. This is because a single non-individual in any of the four roles would lead to the exclusion of the entire observation. As a robustness check, we exclude those repeated sales that involve non-individuals in any of the four roles. Results presented in Table 7 are in line with our baseline results in Table 4; nonwhite move-ins reduce the price appreciation between repeated sales and the racial prejudice effects diminish with distance.

While the results remain largely the same, we do observe slightly stronger evidence of racial prejudice when excluding non-individuals (-0.137 vs. -0.127). In other words, for a typical house of \$380,000, ten extra nonwhite neighbors in the immediate neighborhood are associated with a decrease in home value of \$5,200 (relative to \$4,800 in our main analysis). This finding is consistent with the presence of racial prejudice, as non-individual buyers are less likely than individual buyers to use the property as their primary residence and, therefore,

may care less about the race of their neighbors.

[Table 7 about here.]

4.3.2 Predicting race from name

As we elaborate in Section 3, one of the limitations of our data is the lack of information to identify the race of the seller in the transaction sample. In other words, while we know who moves into the neighborhood, we are not able to identify who moves out of the neighborhood. Therefore, we are not able to pin down the exact change in racial composition of a neighborhood. The racial composition of a neighborhood changes only when a white move-in replaces a nonwhite household or a nonwhite move-in replace a white household. When the buyer and seller share the same race, the neighborhood’s racial composition does not change. To help mitigate this limitation, we apply the *ethnicolr* Python package to predict the seller’s race in the transaction sample based on first and last name. The algorithm follows a model trained on the Florida voter registration database where voters self-report their race; see [Chintalapati et al. \(2023\)](#). When counting the number of white and nonwhite move-ins, we only consider those transactions from a white seller (predicted) to a nonwhite buyer or from a nonwhite seller (predicted) to a white buyer. Given that each nonwhite move-in replaces a white household in the neighborhood and each white move-in replaces a nonwhite household, we define the number of net nonwhite move-ins as the difference between nonwhite and white move-ins. We repeat the analysis using net nonwhite move-ins as the racial composition variable and report the results in Table 8. In line with our baseline results in Table 4, net nonwhite move-ins reduce the price appreciation between repeated sales, which provides evidence of racial prejudice effects. Moreover, racial prejudices diminish with distance.

[Table 8 about here.]

4.4 Do racial preferences drive the effects?

We argue that racial preferences are behind the effects that we have identified in this paper. In this section, through a series of sub-sample analyses, we provide further evidence to show that the racial prejudice effects that we estimate in Section 4.1 are indeed taste-based racial preferences (Becker, 1957).

4.4.1 Market thickness

If what we estimate are racial preferences, then we should expect the preferences to have a smaller influence when the market is “thicker.” When transactions are frequent in a neighborhood, people find it easier to move in or out of there. Someone who has a strong preference against his or her neighbors can move out, while someone who cares less, or even has the opposite preference, may find it attractive to move in. The market force will attenuate the racial prejudice effects, following the logic of Becker (1957). In the top panel of Table 9, we add the interactions of the nonwhite and white racial composition variables with a thick market indicator, which equals one if a neighborhood’s average annual transaction volume is in the top 50% within the census tract and zero otherwise. Consistent with our expectation, we find less prominent racial prejudice effects in thicker markets; nonwhite move-ins have a significantly negative impact on housing price appreciation between repeated sales but the negative impact is attenuated by the thick market indicator. In particular, in the immediate neighborhood of 0.2-mile radius, the marginal effects of nonwhite move-ins are estimated to be -0.204 in non-thick markets and -0.122 in thick markets, corresponding to a decrease in home value of \$7,800 in the former type of markets and \$4,600 in the latter type of markets due to ten extra nonwhite neighbors, for a typical house of \$380,000. This provides the first piece of supporting evidence for the racial prejudice effects that we estimate being driven by taste-based racial preferences.

We also interact the thick market indicator with the race of the buyer and seller in Table 9 to check if the racial price discrimination effects are also driven by racial preferences. However, as the bottom panel of the table shows, the thick market indicator does not significantly

moderate the racial price discrimination effects.

[Table 9 about here.]

4.4.2 Income level

If what we estimate are racial preferences, then we should expect the preferences to be associated with the homebuyer's income level, which could impact racial preferences in two different ways. First, income level is highly correlated with education level and it is well documented in the literature that better educated individuals are less conducive to prejudice toward non-whites.¹⁴ While we do not directly observe the education background of the homebuyers in our dataset, we expect less racial prejudices among higher income homebuyers. Second, income level determines a homebuyer's willingness and/or ability to pay for racial preferences in a marketplace like the housing market.¹⁵ If homebuyers are willing to pay higher prices for white neighbors, we would expect more realized racial prejudices among higher income homebuyers. To test these channels, we split the repeated-sales sample into two sub-samples based on the income level of the homebuyer and define a high income dummy variable as one if the homebuyer's income is in the top 50% among all homebuyers within the same census tract and in the same transaction year, and zero otherwise. We add the interactions of the nonwhite and white racial composition variables with the high income indicator in the top panel of Table 10. As the interaction between nonwhite move-ins and the high income indicator suggests, racial prejudices are stronger among lower income homebuyers and weaker among higher income homebuyers. In particular, in the immediate neighborhood of 0.2-mile radius, the marginal effects of nonwhite move-ins are estimated to be -0.143 in low-income markets and -0.116 in high-income markets, corresponding to a decrease in home value of \$5,400 in the former type of markets and \$4,400 in the latter type of markets due to ten extra nonwhite neighbors, for a

¹⁴Based on the nationwide survey conducted by the National Opinion Research Center, [Maykovich \(1975\)](#) concludes that variables of region, age, and education are found to account for the greatest variance in prejudice.

¹⁵As documented in [Gyourko et al. \(2013\)](#), the size of high income households is useful in explaining geographic differences in house price appreciation.

typical house of \$380,000. This provides the second piece of supporting evidence for the racial prejudice effects that we estimate being driven by taste-based racial preferences.

We also interact the high income indicator with the race of the buyer and seller. As the bottom panel of Table 10 shows, the high income dummy does not significantly moderate the racial price discrimination effects.

[Table 10 about here.]

4.4.3 Political stances

It is often argued that racial preferences are correlated with political stances, and there are studies that try to establish such a relationship based on survey data (Kuziemko and Washington, 2018). However, to our best knowledge, there is no study trying to find the correlation in a market setting that is largely anonymous. Potential buyers who have a preference against some races give a lower offer price or buy from another neighborhood altogether.

We obtain the 2020 presidential election data at the county level from the Virginia Department of Elections and interact nonwhite and white move-ins with the (demeaned) percentage of votes that the Democratic candidate Joe Biden received. As Table 11 shows, there is a negative association between housing price appreciation and nonwhite households that move into the neighborhood. In more democratic counties, however, the racial prejudice effects are smaller.¹⁶ In particular, for a county with an average Biden percentage, an extra nonwhite household in the immediate neighborhood with a radius of 0.2 miles would reduce housing price appreciation between repeated sales by 0.135 percentage points. In other words, in a Virginia county with an average percentage of Democratic votes, the move-ins of ten extra nonwhite households are associated with a decrease in home value of \$5,100 for a typical house of \$380,000. A one standard-deviation (i.e., 0.173) increase in the Biden percentage is associated with a decrease in the racial prejudice effects of about 0.117, which is nearly enough to offset the average distaste

¹⁶As a robustness check, we repeat this analysis using the 2008 presidential election data to ensure exogeneity of voting outcomes, given that our sample spans from 2007 to 2020, and results remain robust.

against nonwhite move-ins. This provides the third piece of supporting evidence for the racial prejudice effects that we estimate being driven by taste-based racial preferences.

We also interact the Biden percentage with the race of the buyer and seller. As the bottom panel of Table 11 shows, the racial price discrimination effects are not moderated by political stances.

[Table 11 about here.]

4.4.4 Public anti-racism statements

While the voting outcomes, to some extent, reflect people's racial preferences, they are not a direct measure of racial preferences. In Table 12, instead of interacting nonwhite and white move-ins with the percentage of votes received by the Democratic presidential candidate, we include their interactions with the (demeaned) BLM score per unit of voting age population, i.e., 18 years and older, at the county level. We find a strong association between racial prejudice effects and the tweeting outcomes; in counties more supportive of Black Lives Matter, racial prejudice effects are significantly less prominent. In particular, for a county with an average BLM score, an extra nonwhite household in the immediate neighborhood with a radius of 0.2 miles would reduce housing price appreciation between repeated sales by 0.103 percentage points. In other words, in a Virginia county with an average BLM score, the move-ins of ten extra nonwhite households are associated with a decrease in home value of \$3,900 for a typical house of \$380,000. A one standard-deviation (i.e., 2.818) increase in the BLM score is associated with a decrease in the racial prejudice effects of about 0.085, which is nearly enough to offset the average distaste against nonwhite move-ins. This provides the fourth piece of supporting evidence for the racial prejudice effects that we estimate being driven by taste-based racial preferences.

We also interact the BLM score with the race of the buyer and seller. As the bottom panel of Table 12 shows, the racial price discrimination effects are not moderated by public anti-racism statements either.

[Table 12 about here.]

It is worth noting that the population base may differ between our voting and tweeting data both in size and composition. While more than half of the U.S. voting age population choose to vote, a little less than 25% of U.S. adults are Twitter users. One major difference between the two variables is that people who tweet are probably (much) younger on average than people who vote. Nevertheless, we find less prominent racial prejudice effects in counties where there is more public anti-racism statements, which further confirms that what we identify are indeed racial prejudice effects.

4.5 Summary and discussion

By relating the price appreciation of a house between its repeated sales to the number of non-white and white households that move into its immediate neighborhood and the race of the buyer and seller, we find strong evidence of both racial prejudice effects and racial price discrimination effects. In particular, our racial prejudice results suggest that as ten extra nonwhite households move into the immediate neighborhood with a radius of 0.2 miles, housing price appreciation between repeated sales decreases by 1.27 percentage points (or \$4,800 for a typical house of \$380,000); this negative marginal impact diminishes with the size of the neighborhood. Our racial price discrimination results suggest that nonwhite sellers receive significantly less (by roughly 1.38% or \$5,200 for a typical house of \$380,000) from selling comparable homes than their white counterparts and pay more (though insignificantly) when buying comparable homes. Our estimates of the racial price discrimination effects are close to what have been found in previous studies, e.g., [Bayer et al. \(2017\)](#), despite the sample difference. These findings remain robust in a series of robustness checks. However, our results differ from those of [Bayer et al. \(2017\)](#) in the sense that the racial price discrimination effects are mainly driven by nonwhite sellers receiving less in our paper but nonwhite buyers paying more in [Bayer et al. \(2017\)](#). One explanation for this difference lies in the sample coverage; our sample spans from 2007 to 2020, a period of declining housing market (buyer's market) in the aftermath of the

global financial crisis, while [Bayer et al. \(2017\)](#)'s sample covers a period of booming housing market (seller's market) between 1990 and 2008 before the financial crisis. In a buyer's market, if being discriminated against, nonwhite buyers can choose to walk away but nonwhite sellers would have to lower their asking price.

By associating the racial prejudice effects with market thickness, income level, political stance, and racial preference made on social media, we find strong evidence indicating that the effects are indeed driven by taste-based racial preferences. However, the same associations do not apply to the racial price discrimination effects, which suggests that the price differentials between nonwhite and white sellers/buyers may not be driven by racial preferences but some other factors that correlate with race, i.e., a form of statistical discrimination ([Phelps, 1972](#); [Arrow, 1972a,b](#)), which has been well documented in the housing market ([Ondrich et al., 2003](#); [Ewens et al., 2014](#); [Gerardi et al., 2023](#)).

5 Heterogeneity of Racial Prejudice Effects

5.1 Differential effects for white and nonwhite homeowners

The baseline results reported in Tables [3](#) and [4](#) shed light on the racial prejudice effects in the housing market and suggest that home values drop (rise) when nonwhite (white) households move into the neighborhood. To study how homeowners are affected by neighbors of the same race differently from those of other races, we interact the number of white and nonwhite move-ins with the race indicator of the buyer and seller. Results are reported in Table [13](#).

[Table 13 about here.]

The coefficients of nonwhite and white move-ins capture the racial prejudice effects for the transactions from white sellers to white buyers, and they remain close to the estimates reported in Table [4](#). Note that white-to-white transactions account for more than 70 percent of the transactions without missing race information. Neither the white-to-nonwhite (W-NW)

indicator nor the nonwhite-to-nonwhite (NW-NW) indicator significantly moderates the negative effects of nonwhite move-ins on housing price appreciation, suggesting no significant difference between white and nonwhite buyers in prejudice against nonwhite neighbors. The interaction between the nonwhite-to-white (NW-W) indicator and nonwhite move-ins is significantly negative for the most immediate neighborhood, which means that white buyers have stronger prejudice against nonwhite neighbors when purchasing from nonwhite sellers.

In our baseline results, homeowners do not significantly benefit from the move-ins of white households. This result remains unchanged in Table 13 for the white-to-white transactions. However, nonwhite sellers significantly benefit from the move-ins of white households no matter the buyer is white or nonwhite. The interaction between nonwhite-to-white and white move-ins and the interaction between nonwhite-to-nonwhite and white move-ins both have a significantly positive impact on housing price appreciation. When white buyers purchase from nonwhite sellers, they are willing to pay a premium of 0.738 percentage points (or \$2,800 for a typical house of \$380,000) for ten additional white neighbors that move into the most immediate neighborhood. Interestingly, when nonwhite buyers purchase from nonwhite sellers, they are willing to pay an equivalent or even slightly higher premium, of about 0.861 percentage points (or \$3,300 for a typical house of \$380,000), for ten additional white neighbors.

The results in Table 13 shed light on the driving factors of racial prejudice effects. First, the distaste against nonwhite neighbors is universal and independent of the race of the seller and buyer. In other words, no matter the buyer is white or nonwhite, they equally view nonwhite neighbors as negative amenities in the neighborhood. This result is likely driven by racial preferences, for which we provide four pieces of supporting evidence in Section 4. Second, the preference for white neighbors differs between white and nonwhite sellers; white neighbors do not necessarily benefit white sellers but significantly benefit nonwhite sellers. One driving factor of this result can be the biased appraisal against owner race and neighborhood race. For example, [Perry et al. \(2018\)](#) find that homes in Black-majority census tracts are devalued by about 23% compared to those in White-majority census tracts, accounting for property and

neighborhood characteristics, and [Ambrose et al. \(2021\)](#) observe systematically lower appraised values for minority-owned homes. Our result suggests that the effects of minority owner and minority neighborhood can reinforce each other; additional white neighbors help appraise a nonwhite owner's home but additional nonwhite neighbors make the appraisal even more downward biased.

5.2 Time-varying effects

According to the General Social Survey, there is evidence that racial preferences are getting weaker over time. For example, one question asks respondents to choose between two hypothetical laws regarding selling a house (the variable name is RACOPEN): "One law says that a homeowner can decide whom to sell their house to, even if that homeowner prefers not to sell to someone because of their race or color. The second law says that a homeowner cannot refuse to sell to someone because of their race or color." The percentage of white respondents who chose the first law has decreased from over 60% in 1973 to 36% in 2004 and 21% in 2018. [Auspurg et al. \(2019\)](#) also document a consistent decline in the extent of discrimination over time. Even though our sample only covers the past decade, we also see some evidence that the racial prejudice effects are decreasing. In Table 14, we interact the move-ins variables with a Post-2017 dummy (where roughly half of the observations have a value of one). We can see that the coefficients on the interaction term are positive and mostly significant, suggesting that the effects, while still present, are weaker in recent years. In particular, for the most immediate neighborhood of a 0.2-mile radius, the marginal effects of nonwhite move-ins are estimated to be -0.165 before 2017 and -0.105 after 2017. These values translate into a decrease in home value of \$6,300 before 2017 and \$4,000 after 2017 due to ten extra nonwhite neighbors, for a typical house of \$380,000. We view it as another supporting evidence that the effects identified in this paper are indeed racial preferences.

[Table 14 about here.]

6 Conclusion

Based on a narrow definition of neighborhoods much smaller than census tracts, we find empirical evidence of taste-based racial prejudices from micro-level housing transaction and mortgage data. Using a narrow geographic definition of neighborhoods allows us to precisely identify relevant neighborhood characteristics that affect house prices. In particular, we find that an additional nonwhite household within a 0.2-mile radius of a repeated transacted house reduces its price appreciation by 1.27 percentage points, which translates into a decrease in home value of \$4,800 due to ten extra nonwhite neighbors for a typical house of \$380,000 in the state of Virginia. While the idea of defining a small neighborhood to study racial prejudice effects coincides with that of [Myers \(2004\)](#), our paper extends her paper with a significantly larger sample and actual prices of repeated sales rather than a survey of owners' self assessment.

The data used in this paper have several limitations, however. First, we use the race of the household that buys a house in the neighborhood to capture changes in its racial composition, but we do not observe the seller's race. Using a race prediction algorithm helps, but the prediction is far from perfect. Second, we do not have information about those who buy a house without a mortgage. We think that transaction data has its strengths and weaknesses, and we are not claiming that our results are necessarily better. We look forward to more research to approach the question with different sources of data.

Despite certain limitations, the data allow us to uncover several new insights. First, homeowners experience a significant financial loss when nonwhite households move into their neighborhoods. We provide compelling evidence that this loss is driven by racial preferences rather than other factors.¹⁷ Second, nonwhite homebuyers are, on average, equally or even slightly more willing than white homebuyers to pay a premium for white neighbors. Third, our analysis reveals that racial prejudices are less pronounced in counties that lean more Democratic in elections and exhibit more anti-racist attitudes on social media.

¹⁷It is worth noting that, despite the analyses that we conduct in Section 4.2 on omitted variables and reverse causality and the use of instrumental variables, it is impossible to eliminate all endogeneity concerns.

References

- Akbar, P. A., Hickly, S. L., Shertzer, A. and Walsh, R. P. (2022). Racial segregation in housing markets and the erosion of black wealth. *The Review of Economics and Statistics*, 1–45.
- Aliprantis, D., Carroll, D. R. and Young, E. R. (2022). What explains neighborhood sorting by income and race?. *FRB of Cleveland Working Paper No. 18-08R*, 1–41.
URL: <https://ssrn.com/abstract=3195529>
- Ambrose, B. W., Conklin, J., Coulson, N. E., Diop, M. and Lopez, L. A. (2021). Do appraiser and borrower race affect valuation?. *Available at SSRN 3951587*, .
- Arrow, K. J. (1972a). Models of job discrimination. in A. H. Pascal (ed.), *Racial Discrimination in Economic Life*. Lexington, MA: D.C. Heath. 83–102.
- Arrow, K. J. (1972b). Some mathematical models of race discrimination in the labor market. in A. H. Pascal (ed.), *Racial Discrimination in Economic Life*. Lexington, MA: D.C. Heath. 187–204.
- Auspurg, K., Schneck, A. and Hinz, T. (2019). Closed doors everywhere? a meta-analysis of field experiments on ethnic discrimination in rental housing markets. *Journal of Ethnic and Migration Studies*, 45(1), 95–114.
- Bakens, J., Florax, R. J. and Mulder, P. (2018). Ethnic drift and white flight: A gravity model of neighborhood formation. *Journal of Regional Science*, 58(5), 921–948.
- Bartik, T. J. (1991). *Who benefits from state and local economic development policies?* Kalamazoo, MI: WE Upjohn Institute for Employment Research.
- Bayer, P., Casey, M., Ferreira, F. and McMillan, R. (2017). Racial and ethnic price differentials in the housing market. *Journal of Urban Economics*, 102, 91–105.
- Bayer, P., Fang, H. and McMillan, R. (2014). Separate when equal? Racial inequality and residential segregation. *Journal of Urban Economics*, 82, 32–48.
- Bayer, P., Ferreira, F. and McMillan, R. (2007). A unified framework for measuring preferences for schools and neighborhoods. *Journal of Political Economy*, 115(4), 588–638.
- Bayer, P., McMillan, R. and Rueben, K. S. (2004). What drives racial segregation? New evidence using census microdata. *Journal of Urban Economics*, 56(3), 514–535.
- Becker, G. S. (1957). *The Economics of Discrimination*. University of Chicago Press.
- Boxell, L. and Conway, J. (2022). Journalist ideology and the production of news: Evidence from movers. *Working Paper*, .
- Boyle, M. and Kiel, K. (2001). A survey of house price hedonic studies of the impact of environmental externalities. *Journal of Real Estate Literature*, 9(2), 117–144.
- Card, D., Mas, A. and Rothstein, J. (2008). Tipping and the dynamics of segregation. *The Quarterly Journal of Economics*, 123(1), 177–218.

- Carrillo, P. E. and Rothbaum, J. L. (2022). Counterfactual dissimilarity: Can changes in demographics and income explain increased racial integration in US cities?. *Journal of Regional Science*, 62(1), 21–56.
- Carrillo, P., Gindelsky, M. and Wentland, S. (2023). When the price is right: Assessing misperceptions in home values and implications for measurement. *Working Paper*, .
- Chambers, D. N. (1992). The racial housing price differential and racially transitional neighborhoods. *Journal of Urban Economics*, 32(2), 214–232.
- Chintalapati, R., Laohaprapanon, S. and Sood, G. (2023). Predicting race and ethnicity from the sequence of characters in a name. *arXiv preprint arXiv:1805.02109*, .
- Crossney, K. B. (2010). Is predatory mortgage lending activity spatially clustered?. *The Professional Geographer*, 62(2), 153–170.
- Cutler, D. M., Glaeser, E. L. and Vigdor, J. L. (1999). The rise and decline of the american ghetto. *Journal of Political Economy*, 107(3), 455–506.
- Daepf, M. I., Hsu, J. W. et al. (2023). The effect of racial composition on neighborhood housing prices: Evidence from hurricane katrina-induced migration. *Journal of Urban Economics*, 134, 103515.
- Ellen, I. G. (2000). Sharing america’s neighborhoods. *Sharing America’s Neighborhoods*. Harvard University Press.
- Ellen, I. G., Horn, K. and O’Regan, K. (2012). Pathways to integration: Examining changes in the prevalence of racially integrated neighborhoods. *Cityscape*, 33–53.
- Ellen, I. G. and Ross, S. L. (2018). Race and the city. *Journal of Housing Economics*, 40, 1–5.
- Ewens, M., Tomlin, B. and Wang, L. C. (2014). Statistical discrimination or prejudice? a large sample field experiment. *Review of Economics and Statistics*, 96(1), 119–134.
- Galster, G. and Godfrey, E. (2005). By words and deeds: Racial steering by real estate agents in the US in 2000. *Journal of the American Planning Association*, 71(3), 251–268.
- Gerardi, K., Willen, P. S. and Zhang, D. H. (2023). Mortgage prepayment, race, and monetary policy. *Journal of Financial Economics*, 147(3), 498–524.
- Glaeser, E. L., Hanushek, E. A. and Quigley, J. M. (2004). Opportunities, race, and urban location: the influence of john kain. *Journal of Urban Economics*, 56(1), 70–79.
- Gonzalez, L. and Ortega, F. (2013). Immigration and housing booms: Evidence from Spain. *Journal of Regional Science*, 53(1), 37–59.
- Gyourko, J., Mayer, C. and Sinai, T. (2013). Superstar cities. *American Economic Journal: Economic Policy*, 5(4), 167–99.

- Howard, G. (2020). The migration accelerator: Labor mobility, housing, and demand. *American Economic Journal: Macroeconomics*, 12(4), 147–179.
- Hwang, J. and Sampson, R. J. (2014). Divergent pathways of gentrification: Racial inequality and the social order of renewal in Chicago neighborhoods. *American Sociological Review*, 79(4), 726–751.
- Ihlanfeldt, K. and Mayock, T. (2009). Price discrimination in the housing market. *Journal of Urban Economics*, 66(2), 125–140.
- Ioannides, Y. M. (2002). Residential neighborhood effects. *Regional Science and Urban Economics*, 32(2), 145–165.
- Ioannides, Y. M. and Zabel, J. E. (2003). Neighbourhood effects and housing demand. *Journal of Applied Econometrics*, 18(5), 563–584.
- Ioannides, Y. M. and Zabel, J. E. (2008). Interactions, neighborhood selection and housing demand. *Journal of Urban Econometrics*, 63(1), 229–252.
- Kahn, M. E. (2021). Racial and ethnic differences in the financial returns to home purchases from 2007 to 2020. *NBER Working Paper 28759*, 1–33.
URL: <http://www.nber.org/papers/w28759>
- Kain, J. F. (1968). Housing segregation, negro employment, and metropolitan decentralization. *The Quarterly Journal of Economics*, 82(2), 175–197.
- Kain, J. F. and Quigley, J. M. (1970). Measuring the value of housing quality. *Journal of the American Statistical Association*, 65(330), 532–548.
- Kain, J. F. and Quigley, J. M. (1972). Housing market discrimination, home-ownership, and savings behavior. *The American Economic Review*, 62(3), 263–277.
- Kain, J. F. and Quigley, J. M. (1975). *Housing Markets and Racial Discrimination: A Microeconomic Analysis*. Columbia University Press.
- Karami, A., Kadari, R. R., Panati, L., Nooli, S. P., Bheemreddy, H. and Bozorgi, P. (2021). Analysis of geotagging behavior: Do geotagged users represent the twitter population?. *ISPRS International Journal of Geo-Information*, 10(6), 373.
- Kermani, A. and Wong, F. (2022). Racial disparities in housing returns. *NBER Working Paper 29306*, 1–100.
URL: <http://www.nber.org/papers/w29306>
- Kiel, K. A. and Zabel, J. E. (1999). The accuracy of owner-provided house values: The 1978–1991 American housing survey. *Real Estate Economics*, 27(2), 263–298.
- Kim, J.-H., Leung, T. C. and Wagman, L. (2017). Can restricting property use be value enhancing? evidence from short-term rental regulation. *Journal of Law and Economics*, 60(2), 309–334.

- Kim, S. (2000). Race and home price appreciation in urban neighborhoods: Evidence from Milwaukee, Wisconsin. *The Review of Black Political Economy*, 28(2), 9–28.
- King, A. T. and Mieszkowski, P. (1973). Racial discrimination, segregation, and the price of housing. *Journal of Political Economy*, 81(3), 590–606.
- Kollmann, T., Marsiglio, S. and Suardi, S. (2018). Racial segregation in the united states since the great depression: A dynamic segregation approach. *Journal of Housing Economics*, 40, 95–116.
- Kuziemko, I. and Washington, E. (2018). Why did the democrats lose the south? bringing new data to an old debate. *American Economic Review*, 108(10), 2830–67.
- Leung, T. C. and Strumpf, K. (2022). All the headlines that are fit to change. *SSRN Working Paper No. 4075398*, 1–40.
URL: <https://ssrn.com/abstract=4075398>
- Leung, T. C. and Tsang, K. P. (2012). Love thy neighbor: Income distribution and housing preferences. *Journal of Housing Economics*, 21(4), 322–335.
- Macpherson, D. A. and Sirmans, G. S. (2001). Neighborhood diversity and house-price appreciation. *The Journal of Real Estate Finance and Economics*, 22(1), 81–97.
- Massey, D. and Denton, N. A. (1993). *American Apartheid: Segregation and the Making of the Underclass*. Harvard University Press.
- Maykovich, M. K. (1975). Correlates of racial prejudice.. *Journal of Personality and Social Psychology*, 32(6), 1014.
- Moye, R. (2014). Neighborhood racial–ethnic change and home value appreciation: Evidence from Philadelphia. *Urban Geography*, 35(2), 236–263.
- Myers, C. K. (2004). Discrimination and neighborhood effects: Understanding racial differentials in US housing prices. *Journal of Urban Economics*, 56(2), 279–302.
- Ondrich, J., Ross, S. and Yinger, J. (2003). Now you see it, now you don't: why do real estate agents withhold available houses from black customers?. *Review of Economics and Statistics*, 85(4), 854–873.
- Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2), 187–204.
- Perry, A., Rothwell, J. and Harshbarger, D. (2018). The devaluation of assets in black neighborhoods. *Brookings*, November, 1–28.
- Phelps, E. S. (1972). The statistical theory of racism and sexism. *The American Economic Review*, 62(4), 659–661.
- Reibel, M. (2000). Geographic variation in mortgage discrimination: Evidence from Los Angeles. *Urban Geography*, 21(1), 45–60.

- Ross, M. H. (1971). Prices, segregation, and racial harmony. *Journal of Black Studies*, 2(2), 225–244.
- Rossi-Hansberg, E., Sarte, P. and Owens III, R. (2010). Housing externalities. *Journal of Political Economy*, 118(3), 485–535.
- Saiz, A. (2003). Room in the kitchen for the melting pot: Immigration and rental prices. *Review of Economics and Statistics*, 85(3), 502–521.
- Saiz, A. and Wachter, S. (2011). Immigration and the neighborhood. *American Economic Journal: Economic Policy*, 3(2), 169–88.
- Schafer, R. (1979). Racial discrimination in the Boston housing market. *Journal of Urban Economics*, 6(2), 176–196.
- Shertzer, A., Twinam, T. and Walsh, R. P. (2016). Race, ethnicity, and discriminatory zoning. *American Economic Journal: Applied Economics*, 8(3), 217–246.
- Sloan, L. and Morgan, J. (2015). Who tweets with their location? Understanding the relationship between demographic characteristics and the use of geoservices and geotagging on Twitter. *PLoS One*, 10(11), e0142209.
- Sloan, L., Morgan, J., Housley, W., Williams, M., Edwards, A., Burnap, P. and Rana, O. (2013). Knowing the tweeters: Deriving sociologically relevant demographics from Twitter. *Sociological Research Online*, 18(3), 74–84.
- Squires, G. D. (1994). *Capital and Communities in Black and White: The Intersections of Race, Class, and Uneven Development*. Albany: State University of New York Press.
- Yinger, J. (1978). The black-white price differential in housing: Some further evidence. *Land Economics*, 54(2), 187–206.

Appendix: Transaction Sample vs. Repeated-Sales Sample

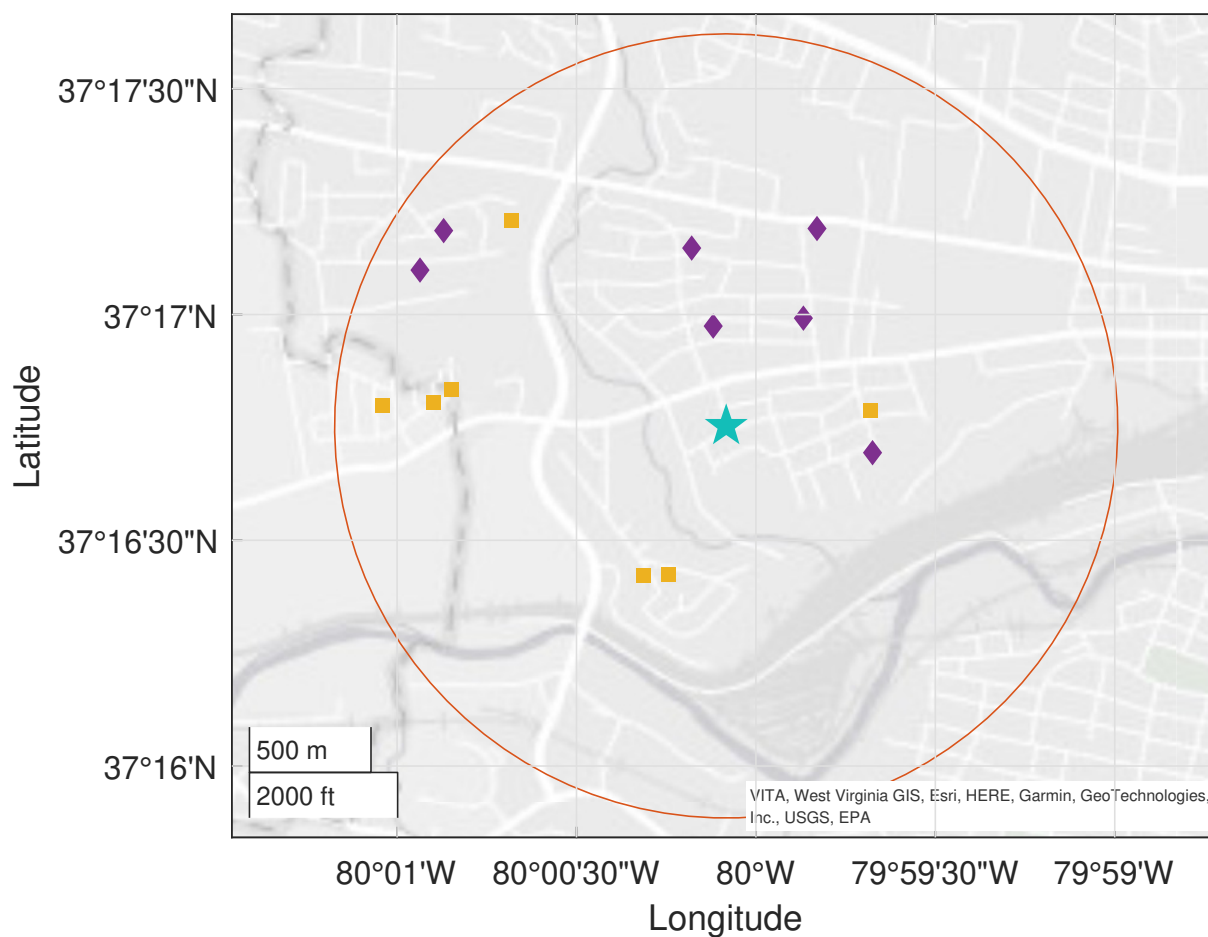
This table performs a two-sample t-test on sale price and applicant income.

[Table 15 about here.]

Appendix: First Stage Results

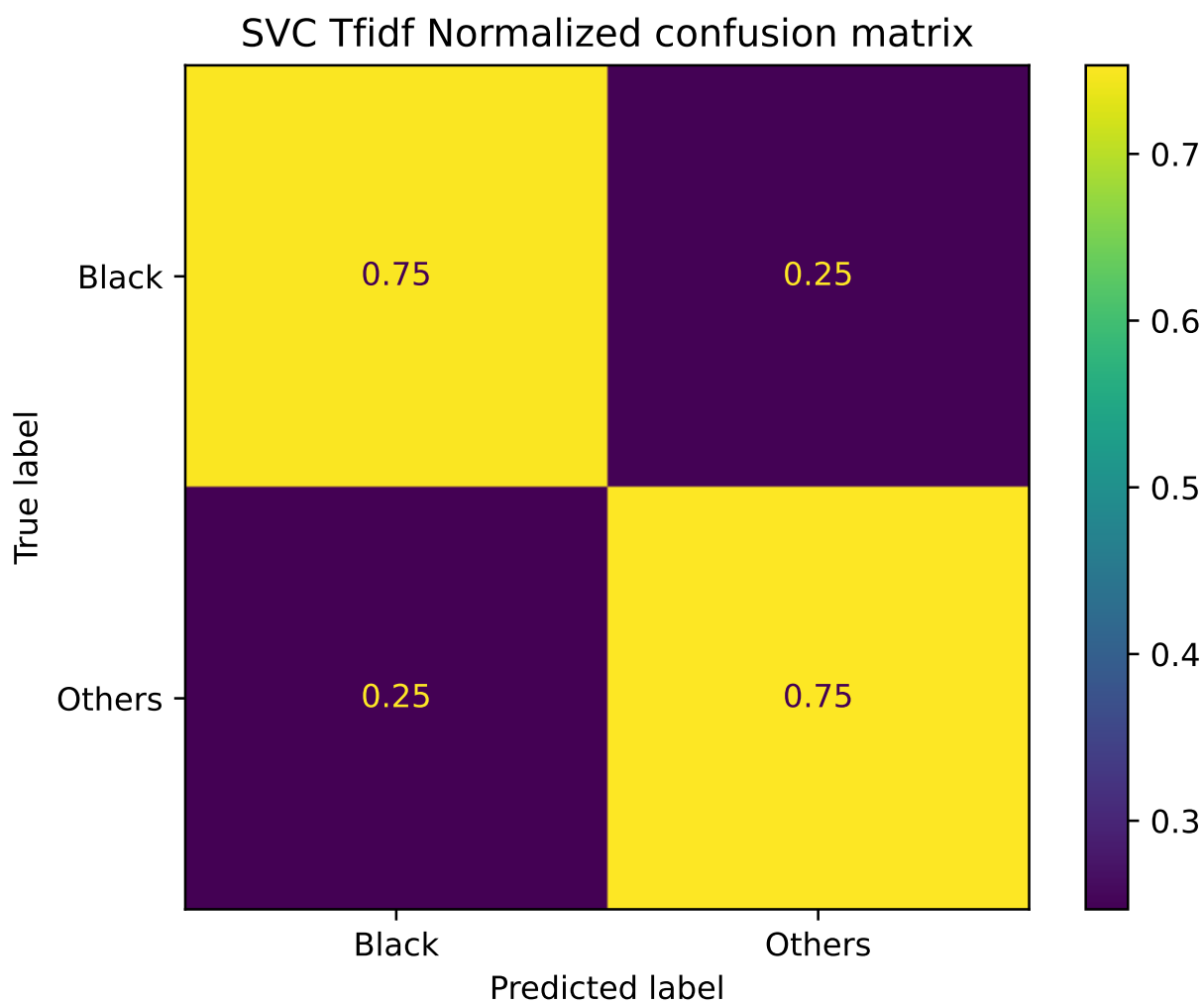
This section presents the first stage regression results of IV estimation. Nonwhite and white move-ins are treated endogenous here. The first set of instruments are nonwhite and white move-ins within the broader 1-mile radius. The second set of instruments are nonwhite and white move-ins in nearby neighborhoods located in the same census tract.

[Table 16 about here.]



Note: The pentagram denotes a property that has been transacted repeatedly. The squares and diamonds denote white and nonwhite households, respectively, that move into the neighborhood of a radius of one mile between the repeated sales.

Figure 1: Illustrative map of one neighborhood



Note: The confusion matrix depicts the percentages of type I and type II errors in the machine learning prediction. “Black” denotes the tweets with hashtag #BlackLivesMatter, and “Others” denotes the tweets with other hashtags (#AllLivesMatter and #BlueLivesMatter).

Figure 2: Confusion matrix of machine learning prediction

Table 1: Summary statistics

Transaction sample	Obs	Mean	Std. Dev.	Min	Max
Sale price (in millions)	591,434	0.358	0.234	0.060	1.336
White	405,666	0.347	0.227	0.060	1.336
Nonwhite	108,802	0.355	0.234	0.060	1.336
Missing race info	76,966	0.420	0.263	0.060	1.336
Applicant income (in millions)	587,490	0.107	0.074	0.023	0.434
White	402,767	0.105	0.073	0.023	0.434
Nonwhite	108,262	0.101	0.069	0.023	0.434
Missing race info	76,461	0.124	0.083	0.023	0.434
Repeated-sales sample	Obs	Mean	Std. Dev.	Min	Max
$\Delta \ln(\text{Price}) \times 100$	95,967	12.758	25.603	-62.527	130.258
Nonwhite move-ins within 0.2 miles	95,967	3.984	6.322	0	111
Nonwhite move-ins within 0.4 miles	95,967	11.753	16.317	0	212
Nonwhite move-ins within 0.6 miles	95,967	22.299	29.529	0	316
Nonwhite move-ins within 0.8 miles	95,967	35.337	45.544	0	463
Nonwhite move-ins within 1.0 mile	95,967	50.557	64.066	0	610
White move-ins within 0.2 miles	95,967	14.071	16.705	0	198
White move-ins within 0.4 miles	95,967	42.709	49.090	0	658
White move-ins within 0.6 miles	95,967	81.593	92.656	0	1112
White move-ins within 0.8 miles	95,967	128.933	144.528	0	1629
White move-ins within 1.0 mile	95,967	183.481	202.951	0	2237
Avg income of move-ins within 0.2 miles	83,832	0.103	0.052	0.023	0.434
Avg income of move-ins within 0.4 miles	86,258	0.104	0.049	0.023	0.434
Avg income of move-ins within 0.6 miles	87,006	0.104	0.048	0.023	0.434
Avg income of move-ins within 0.8 miles	87,325	0.104	0.046	0.023	0.434
Avg income of move-ins within 1.0 miles	87,534	0.104	0.045	0.023	0.434
$\Delta(\text{Buyer income})$	94,600	0.007	0.063	-0.368	0.387
$\Delta(\text{Buyer income})^2$	94,600	0.002	0.025	-0.155	0.175
$\Delta(\text{Buyer DTI})$	94,600	0.169	1.583	-22.402	35.368
$\Delta(\text{Buyer DTI})^2$	94,600	1.427	16.309	-663.054	1401.705
$\Delta \text{NonwhiteSeller}$	Frequency	Percent			
-1	809	10.24			
0	6,290	79.61			
1	802	10.15			
$\Delta \text{NonwhiteBuyer}$	Frequency	Percent			
-1	6,758	9.20			
0	56,472	76.89			
1	10,217	13.91			

Price, income, and loan amount (for the DTI calculation) are winsorized at 1% and 99% quantile respectively.

Table 2: Summary statistics of voting and tweeting outcomes

	Obs	Mean	Std. Dev.	Min	Max
Biden pct	110	0.437	0.173	0.150	0.829
BLM score	109	3.033	2.818	0.187	14.373

Both Biden pct and BLM score are at the county level, with a correlation coefficient of 0.65.

Table 3: Baseline racial prejudice effects

	(1)	(2)	(3)	(4)
Racial composition				
Nonwhite move-ins	-0.179*** (-7.15)	-0.181*** (-7.21)	-0.163*** (-6.61)	-0.127*** (-5.81)
White move-ins	0.00409 (0.34)	0.00386 (0.32)	0.0174 (1.50)	0.0128 (1.20)
Race of the buyer and seller				
Δ NonwhiteSeller		-1.606** (-2.98)	-1.696** (-3.20)	-1.379** (-2.69)
Δ NonwhiteBuyer		0.362 (1.78)	0.404* (2.02)	0.112 (0.61)
Control variables				
Avg income of move-ins			-24.78*** (-6.10)	-19.42*** (-5.16)
Δ (Buyer income)				385.0*** (30.14)
Δ (Buyer income) ²				-595.3*** (-24.11)
Δ (Buyer DTI)				9.700*** (20.24)
Δ (Buyer DTI) ²				-0.337*** (-7.64)
Constant	13.62*** (81.89)	10.50*** (35.15)	11.60*** (20.29)	8.677*** (15.73)
Observations	89642	89642	89642	89642
Adjusted R^2	0.328	0.330	0.344	0.414

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Non-white move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within 0.2 miles from the repeatedly transacted house between its repeated sales. All regressions include census tract by year of first sale fixed effects and census tract by year of second sale fixed effects, as well as three missing value indicators for Δ NonwhiteSeller, Δ NonwhiteBuyer, and Avg income of move-ins. Each missing value indicator equals one if the corresponding variable (which is later reset to zero) is missing and zero otherwise. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 4: Diminishing racial prejudice effects with distance

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.127*** (-5.81)	-0.0496*** (-4.55)	-0.0321*** (-4.21)	-0.0156** (-2.60)	-0.00843 (-1.67)
White move-ins	0.0128 (1.20)	0.00131 (0.25)	0.00376 (1.05)	0.00334 (1.18)	0.00346 (1.44)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.379** (-2.69)	-1.343** (-2.63)	-1.389** (-2.72)	-1.379** (-2.70)	-1.377** (-2.69)
Δ NonwhiteBuyer	0.112 (0.61)	0.131 (0.72)	0.138 (0.76)	0.134 (0.74)	0.133 (0.74)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.414	0.416	0.418	0.418	0.419
δ	1.785	5.595	5.611	8.787	

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The δ estimate measures the proportional bias due to unobservables that would have to exist to drive the coefficient of nonwhite move-ins to zero as in [Oster \(2019\)](#).

Table 5: Reverse causality test

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Annual price appreciation (in percent)	-0.00529 (-1.55)	-0.00163 (-0.54)	-0.000789 (-0.29)	-0.00298 (-1.22)	-0.00243 (-1.17)
Zero move-ins indicator	-31.52*** (-60.12)	-21.79*** (-34.42)	-19.47*** (-29.37)	-18.53*** (-29.81)	-17.64*** (-31.34)
Observations	91572	91572	91572	91572	91572
Adjusted R^2	0.758	0.792	0.852	0.892	0.917

The dependent variable is the share of nonwhite move-ins (in percent) within one year after the second sale of the house. The independent variable of interest is the annual price appreciation (in percent) between repeated sales. The zero move-ins indicator equals one if there is zero white or zero nonwhite move-ins within one year after the second sale and zero when there is at least one white and one nonwhite move-ins. When the indicator equals one, the dependent variable is set to zero. All regressions include census tract by year of first sale fixed effects and census tract by year of second sale fixed effects. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6: Racial prejudice effects using instrumental variables

	(1)	(2)	(3)	(4)	
Distance (miles)	0.2	0.4	0.6	0.8	
Racial composition					
Nonwhite move-ins	-0.137 ^{\$} (-1.90)	-0.0406 ^{\$} (-1.92)	-0.0194 ^{\$} (-1.81)	-0.0117 ^{\$} (-1.73)	
White move-ins	0.0693 (1.59)	0.0141 (1.20)	0.00714 (1.32)	0.00467 (1.44)	
Race of the buyer and seller					
Δ NonwhiteSeller	-1.366** (-2.66)	-1.336** (-2.61)	-1.383** (-2.71)	-1.378** (-2.69)	
Δ NonwhiteBuyer	0.118 (0.65)	0.132 (0.73)	0.139 (0.77)	0.134 (0.74)	
Observations	89642	89642	89642	89642	
Adjusted R^2	0.129	0.132	0.134	0.135	
	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.179* (-1.99)	-0.0686 (-1.36)	-0.0325 (-0.98)	-0.0132 (-0.55)	-0.00380 (-0.21)
White move-ins	0.0737 ^{\$} (1.83)	0.0194 (0.99)	0.0176 (1.32)	0.0130 (1.41)	0.0104 (1.48)
Race of the buyer and seller					
Δ NonwhiteSeller	-0.461 (-0.86)	-0.428 (-0.80)	-0.475 (-0.89)	-0.472 (-0.88)	-0.488 (-0.91)
Δ NonwhiteBuyer	0.160 (0.85)	0.165 (0.89)	0.173 (0.93)	0.165 (0.88)	0.162 (0.87)
Observations	72732	72732	72732	72732	72732
Adjusted R^2	0.114	0.116	0.117	0.118	0.118

In the top panel, the instruments for nonwhite and white move-ins within a certain distance are the nonwhite and white move-ins within the broader 1-mile radius of the same repeatedly transacted property. In the bottom panel, the instruments for nonwhite and white move-ins within a certain distance are the average nonwhite and white move-ins within the same distance of other repeatedly transacted properties located in the same census tract between the same two years of repeated transactions. All regressions include the most comprehensive set of control variables as in Table 3. Standard errors are clustered at the census tract level. t statistics in parentheses. ^{\$} $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 7: Racial prejudice effects excluding non-individuals

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.137*** (-4.14)	-0.0469** (-2.72)	-0.0310* (-2.51)	-0.0180 (-1.84)	-0.0103 (-1.24)
White move-ins	-0.0239 (-1.55)	-0.00807 (-1.05)	-0.00230 (-0.43)	0.000411 (0.10)	0.000999 (0.30)
Race of the buyer and seller					
Δ NonwhiteSeller	-0.881 (-1.43)	-0.906 (-1.47)	-0.952 (-1.54)	-0.937 (-1.52)	-0.938 (-1.52)
Δ NonwhiteBuyer	0.141 (0.60)	0.131 (0.55)	0.133 (0.56)	0.139 (0.58)	0.134 (0.56)
Observations	35179	35179	35179	35179	35179
Adjusted R^2	0.274	0.272	0.271	0.270	0.269

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Non-white move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 8: Racial prejudice effects using net nonwhite move-ins

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Net nonwhite move-ins	-0.175* (-2.52)	-0.0730** (-3.11)	-0.0387** (-3.13)	-0.0224** (-2.69)	-0.0157* (-2.54)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.290* (-2.47)	-1.289* (-2.47)	-1.285* (-2.47)	-1.286* (-2.47)	-1.290* (-2.48)
Δ NonwhiteBuyer	0.0783 (0.42)	0.0799 (0.43)	0.0823 (0.45)	0.0816 (0.44)	0.0816 (0.44)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.403	0.403	0.403	0.403	0.403

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Non-white move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. Net nonwhite move-ins is the difference between nonwhite and white move-ins. All regressions include the most comprehensive set of control variables as in Table 3. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 9: Racial prejudice effects and market thickness

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.204*** (-3.95)	-0.153*** (-7.12)	-0.0899*** (-5.89)	-0.0450*** (-4.02)	-0.0317*** (-3.60)
Nonwhite move-ins $\times$ Thick market	0.0818 (1.82)	0.0993*** (5.96)	0.0537*** (4.87)	0.0281*** (3.95)	0.0228*** (4.57)
White move-ins	-0.00255 (-0.10)	-0.00146 (-0.15)	0.00804 (1.27)	0.00439 (0.95)	0.00366 (0.98)
White move-ins $\times$ Thick market	0.0183 (0.92)	0.00499 (0.88)	-0.00145 (-0.42)	0.00136 (0.59)	0.00230 (1.44)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.997* (-2.47)	-1.840* (-2.33)	-2.183** (-2.79)	-2.027* (-2.53)	-2.095** (-2.61)
Δ NonwhiteSeller $\times$ Thick market	1.219 (1.16)	0.969 (0.92)	1.567 (1.44)	1.216 (1.13)	1.294 (1.20)
Δ NonwhiteBuyer	-0.134 (-0.46)	-0.0929 (-0.33)	0.194 (0.68)	0.268 (0.95)	0.337 (1.16)
Δ NonwhiteBuyer $\times$ Thick market	0.482 (1.37)	0.442 (1.30)	-0.118 (-0.35)	-0.255 (-0.77)	-0.394 (-1.15)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.415	0.417	0.418	0.419	0.420

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Thick market equals one if a neighborhood's average annual transaction volume is in the top 50% within the census tract and zero otherwise. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and the thick market indicator. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 10: Racial prejudice effects and income level

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.143*** (-5.84)	-0.0622*** (-5.33)	-0.0409*** (-5.21)	-0.0217*** (-3.53)	-0.0130* (-2.53)
Nonwhite move-ins $\times$ High income	0.0274 (1.19)	0.0229* (2.21)	0.0167** (2.87)	0.0123** (3.12)	0.00958*** (3.35)
White move-ins	0.0162 (1.45)	0.00256 (0.49)	0.00425 (1.17)	0.00350 (1.20)	0.00348 (1.41)
White move-ins $\times$ High income	-0.00910 (-0.94)	-0.00334 (-0.96)	-0.00148 (-0.77)	-0.000696 (-0.55)	-0.000320 (-0.35)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.385* (-2.01)	-1.321 (-1.93)	-1.352* (-1.97)	-1.342 (-1.96)	-1.346 (-1.96)
Δ NonwhiteSeller $\times$ High income	-0.00419 (-0.00)	-0.0700 (-0.08)	-0.100 (-0.11)	-0.102 (-0.11)	-0.0940 (-0.10)
Δ NonwhiteBuyer	-0.00301 (-0.01)	0.0466 (0.21)	0.0615 (0.28)	0.0613 (0.27)	0.0622 (0.28)
Δ NonwhiteBuyer $\times$ High income	0.219 (0.64)	0.157 (0.46)	0.141 (0.42)	0.131 (0.39)	0.127 (0.38)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.415	0.417	0.418	0.419	0.419

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. High income equals one if the homebuyer is in the top 50% among all homebuyers within the census tract - transaction year and zero otherwise. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and the high income indicator. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 11: Racial prejudice effects and voting outcomes

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.135*** (-6.39)	-0.0649*** (-6.01)	-0.0523*** (-6.32)	-0.0327*** (-4.49)	-0.0215*** (-3.50)
Nonwhite move-ins $\times$ Biden pct	0.674** (2.78)	0.640*** (4.52)	0.607*** (5.66)	0.468*** (5.02)	0.358*** (4.36)
White move-ins	0.0189 (1.76)	0.00597 (1.15)	0.00863* (2.32)	0.00715* (2.43)	0.00637** (2.62)
White move-ins $\times$ Biden pct	0.124 (1.43)	0.0129 (0.37)	-0.0132 (-0.56)	-0.0102 (-0.53)	0.000632 (0.04)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.348** (-2.67)	-1.295* (-2.58)	-1.338** (-2.66)	-1.336** (-2.65)	-1.344** (-2.67)
Δ NonwhiteSeller $\times$ Biden pct	-2.528 (-0.48)	-2.680 (-0.51)	-2.404 (-0.46)	-2.196 (-0.42)	-2.379 (-0.46)
Δ NonwhiteBuyer	0.130 (0.72)	0.155 (0.86)	0.160 (0.89)	0.158 (0.88)	0.155 (0.87)
Δ NonwhiteBuyer $\times$ Biden pct	-1.402 (-0.73)	-1.444 (-0.76)	-1.483 (-0.78)	-1.433 (-0.75)	-1.461 (-0.76)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.416	0.418	0.419	0.420	0.420

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Biden pct is the (demeaned) percentage of votes at the county level received by the Democratic presidential candidate Joe Biden in the 2020 election. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and the Biden percentage. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 12: Racial prejudice effects and tweeting outcomes

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.103*** (-4.85)	-0.0332** (-3.17)	-0.0199** (-2.78)	-0.00672 (-1.23)	-0.00208 (-0.44)
Nonwhite move-ins $\times$ BLM score	0.0300*** (3.59)	0.0210*** (4.51)	0.0161*** (4.45)	0.0142*** (4.69)	0.0108*** (4.02)
White move-ins	0.0132 (1.27)	0.00127 (0.24)	0.00394 (1.09)	0.00349 (1.22)	0.00327 (1.37)
White move-ins $\times$ BLM score	0.00213 (0.74)	-0.000332 (-0.26)	-0.000494 (-0.56)	-0.000589 (-0.80)	-0.0000849 (-0.13)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.415** (-2.74)	-1.377** (-2.67)	-1.410** (-2.74)	-1.400** (-2.72)	-1.406** (-2.74)
Δ NonwhiteSeller $\times$ BLM score	-0.179 (-1.05)	-0.182 (-1.07)	-0.171 (-1.00)	-0.168 (-0.99)	-0.167 (-0.99)
Δ NonwhiteBuyer	0.107 (0.58)	0.127 (0.69)	0.133 (0.73)	0.126 (0.69)	0.120 (0.66)
Δ NonwhiteBuyer $\times$ BLM score	-0.0415 (-0.69)	-0.0471 (-0.80)	-0.0492 (-0.83)	-0.0489 (-0.83)	-0.0502 (-0.85)
Observations	89635	89635	89635	89635	89635
Adjusted R^2	0.415	0.417	0.418	0.419	0.420

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. BLM score is the (demeaned) Twitter BLM measure divided by voting age population (in thousands) at the county level. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and the BLM score. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 13: Differential effects for white and nonwhite homeowners

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.107*** (-3.91)	-0.0391** (-3.05)	-0.0276*** (-3.30)	-0.0119 (-1.91)	-0.00646 (-1.24)
W-NW $\times$ Nonwhite move-ins	-0.0221 (-0.69)	-0.0109 (-0.82)	-0.00216 (-0.28)	-0.00253 (-0.48)	-0.000843 (-0.22)
NW-W $\times$ Nonwhite move-ins	-0.107** (-2.73)	-0.0257 (-1.60)	-0.0129 (-1.38)	-0.0103 (-1.61)	-0.00779 (-1.65)
NW-NW $\times$ Nonwhite move-ins	-0.0340 (-0.96)	-0.0193 (-1.27)	-0.00458 (-0.51)	-0.00380 (-0.61)	-0.00259 (-0.55)
White move-ins	0.00125 (0.11)	-0.00246 (-0.46)	0.00188 (0.51)	0.00210 (0.74)	0.00261 (1.08)
W-NW $\times$ White move-ins	0.00427 (0.29)	-0.000124 (-0.02)	-0.00198 (-0.62)	-0.00145 (-0.69)	-0.00106 (-0.69)
NW-W $\times$ White move-ins	0.0738*** (3.32)	0.0260** (3.27)	0.0143*** (3.39)	0.00919*** (3.47)	0.00657*** (3.55)
NW-NW $\times$ White move-ins	0.0861** (3.15)	0.0259* (2.57)	0.00951 (1.65)	0.00461 (1.25)	0.00292 (1.13)
Race of the buyer and seller					
Δ NonwhiteSeller	-1.476** (-2.78)	-1.402** (-2.66)	-1.423** (-2.70)	-1.407** (-2.67)	-1.411** (-2.68)
Δ NonwhiteBuyer	1.582** (3.15)	1.873*** (3.56)	1.857*** (3.43)	1.730** (3.13)	1.560** (2.81)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.416	0.418	0.419	0.420	0.420

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. W-NW, NW-W, and NW-NW stand for white-to-nonwhite, nonwhite-to-white, and nonwhite-to-nonwhite transactions, while the base group is defined as white-to-white transactions. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and seller-buyer race. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 14: Racial prejudice effects over time

	(1)	(2)	(3)	(4)	(5)
Distance (miles)	0.2	0.4	0.6	0.8	1.0
Racial composition					
Nonwhite move-ins	-0.165*** (-5.27)	-0.0873*** (-5.32)	-0.0624*** (-5.34)	-0.0402*** (-4.38)	-0.0308*** (-4.21)
Nonwhite move-ins $\times$ Post2017	0.0605 (1.71)	0.0551** (2.97)	0.0430*** (3.38)	0.0334*** (3.49)	0.0298*** (4.03)
White move-ins	0.00991 (0.68)	0.00354 (0.52)	0.00534 (1.19)	0.00456 (1.33)	0.00487 (1.77)
White move-ins $\times$ Post2017	0.000112 (0.01)	-0.00462 (-0.70)	-0.00309 (-0.76)	-0.00224 (-0.80)	-0.00252 (-1.25)
Race of the buyer and seller					
Δ NonwhiteSeller	-2.216* (-2.21)	-2.225* (-2.24)	-2.312* (-2.34)	-2.363* (-2.38)	-2.358* (-2.37)
Δ NonwhiteSeller $\times$ Post2017	1.199 (1.03)	1.276 (1.10)	1.339 (1.16)	1.421 (1.23)	1.405 (1.21)
Δ NonwhiteBuyer	0.308 (1.07)	0.336 (1.18)	0.331 (1.17)	0.324 (1.15)	0.326 (1.15)
Δ NonwhiteBuyer $\times$ Post2017	-0.359 (-1.00)	-0.374 (-1.05)	-0.353 (-0.99)	-0.351 (-0.98)	-0.353 (-0.99)
Observations	89642	89642	89642	89642	89642
Adjusted R^2	0.415	0.417	0.418	0.419	0.419

The dependent variable is the nominal price appreciation, in percent, between repeated sales of a house. Post-2017 equals one if the second sale of the house happens after 2017 and zero otherwise. Nonwhite move-ins and white move-ins are the number of nonwhite and white homebuyers, respectively, within a certain distance from the repeatedly transacted house between its repeated sales. All regressions include the most comprehensive set of control variables as in Table 3 and the interaction between average income of move-ins and the Post-2017 dummy variable. Standard errors are clustered at the census tract level. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 15: Transaction sample vs. repeated-sales sample comparison

Sale price (in millions)					
Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]
0	408,730	0.3636	0.0004	0.2428	0.3629 – 0.3644
1	182,704	0.3455	0.0005	0.2137	0.3445 – 0.3464
Combined	591,434	0.3580	0.0003	0.2343	0.3574 – 0.3586
diff		0.0181	0.0006		0.0169 – 0.0194
Applicant income (in millions)					
Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]
0	406,116	0.1080	0.0001	0.0756	0.1078 – 0.1083
1	181,374	0.1050	0.0002	0.0705	0.1047 – 0.1054
Combined	587,490	0.1071	0.0001	0.0741	0.1069 – 0.1073
diff		0.0030	0.0002		0.0026 – 0.0034

Group 0 comprises all properties in the transaction sample that are not included in the repeated-sales sample, while Group 1 consists of all properties within the repeated-sales sample.

Table 16: First stage regression results of IV estimation

Distance (miles)	0.2		0.4		0.6		0.8	
	NW	W	NW	W	NW	W	NW	W
Nonwhite moveins IV	0.0719*** (9.50)	0.0213** (3.00)	0.234*** (16.18)	0.0412* (2.23)	0.462*** (26.24)	0.0350 (1.48)	0.743*** (57.73)	0.0234 (1.40)
White moveins IV	-0.00577*** (-3.65)	0.0469*** (11.59)	-0.0106*** (-3.48)	0.188*** (17.51)	-0.0112** (-2.96)	0.427*** (29.45)	-0.00653* (-2.11)	0.726*** (65.01)
ΔNonwhiteSeller	-0.0544 (-0.57)	-0.323 ^{\$} (-1.68)	-0.162 (-0.87)	-0.823* (-2.00)	-0.430* (-2.04)	-1.148* (-2.21)	-0.501* (-2.45)	-1.339** (-2.90)
ΔNonwhiteBuyer	-0.0610 (-1.50)	-0.144 ^{\$} (-1.69)	-0.0206 (-0.27)	-0.147 (-0.84)	-0.0379 (-0.40)	-0.222 (-1.07)	-0.0596 (-0.76)	-0.0753 (-0.45)
Observations	89642	89642	89642	89642	89642	89642	89642	89642
Adjusted R ²	0.627	0.727	0.816	0.868	0.917	0.943	0.975	0.985

Distance (miles)	0.2		0.4		0.6		0.8		1.0	
	NW	W	NW	W	NW	W	NW	W	NW	W
Nonwhite moveins IV	-0.360*** (-9.82)	0.0646* (2.32)	-0.358*** (-9.29)	0.0304 (0.86)	-0.360*** (-9.25)	0.0181 (0.52)	-0.382*** (-10.32)	0.0376 (0.97)	-0.405*** (-11.60)	0.0548 (1.28)
White moveins IV	0.00949 (1.07)	-0.374*** (-12.79)	-0.00237 (-0.30)	-0.391*** (-18.17)	0.00175 (0.24)	-0.362*** (-14.33)	0.00586 (0.83)	-0.377*** (-14.40)	0.0106 (1.54)	-0.384*** (-14.28)
ΔNonwhiteSeller	-0.0998 (-0.89)	-0.149 (-0.72)	-0.186 (-0.84)	-0.391 (-0.85)	-0.346 (-1.08)	-0.102 (-0.15)	-0.326 (-0.77)	0.132 (0.15)	0.237 (0.46)	1.843 ^{\$} (1.70)
ΔNonwhiteBuyer	-0.0521 (-1.05)	-0.106 (-1.04)	-0.0827 (-0.83)	-0.0739 (-0.34)	-0.128 (-0.89)	-0.000153 (-0.00)	-0.170 (-0.95)	0.316 (0.77)	-0.115 (-0.59)	0.460 (0.93)
Observations	72732	72732	72732	72732	72732	72732	72732	72732	72732	72732
Adjusted R ²	0.593	0.694	0.748	0.820	0.831	0.881	0.879	0.915	0.911	0.933

The dependent variable is the nonwhite (NW) or white (W) move-ins within a certain distance of a repeatedly transacted property. In the top panel, the instruments for nonwhite and white move-ins within a certain distance are the nonwhite and white move-ins within the broader 1-mile radius of the same repeatedly transacted property. In the bottom panel, the instruments for nonwhite and white move-ins within a certain distance are the average nonwhite and white move-ins within the same distance of other repeatedly transacted properties located in the same census tract between the same two years of repeated transactions. All regressions include the most comprehensive set of control variables as in Table 3. Standard errors are clustered at the census tract level. t statistics in parentheses. ^{\$} $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.